

Recursive Bayesian Compression and False Certainty*

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Abstract

A fully Bayesian learner retains everything that past observations imply about an uncertain state. We study a learner who instead keeps only separate assessments of two components of the state: after each new signal, she updates correctly, records the two posterior marginals, discards their association, and before the next signal reconstructs a joint belief by treating those marginals as independent. Thus every signal is observed and the statistical model is correct, but information created by earlier updating can be lost between periods. We characterize when this repeated compression can lead the learner to become certain of a false state. A state can attract learning when it explains the data better than every alternative obtained by changing only one of its components, even though another state provides a better overall explanation. In a binary environment, we show that this mechanism is robust: throughout an open set of correctly specified models, the compressed learner converges to certainty in a particular false state with positive probability. Finally, we characterize when marginal records lose no information relevant for future learning: this happens exactly when each signal affects the two state components in a multiplicatively separable way, so that Bayesian updating never creates association between them. With two binary components, storing one additional measure of association is enough to restore exact Bayesian updating.

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1 Introduction

Economic learning often concerns several related dimensions of an uncertain state. A loan officer may assess both a borrower’s underlying quality and whether recent losses reflect growth or solvency risk. A hiring committee may assess both ability and organizational fit. After each new observation, a fully Bayesian learner can retain the complete history of observations, or equivalently the full joint belief that summarizes everything the history implies about the two dimensions, including how they are associated.

Actual records often retain less. Rather than storing the full joint belief, an organization may record only the marginal assessment of each dimension: one probability for underlying quality and another for the source of recent losses, or one score for ability and another for organizational fit. We call these marginal records. They preserve what the learner believes about each component separately, but not what the learner believes about their association. Two complete histories can therefore generate the same marginal record even though they imply different joint beliefs.

This paper studies learning when marginal records, rather than complete histories or full joint beliefs, are carried forward across dates. The distinction matters because marginal records do not simply summarize the current assessment; they become the input into the next round of updating. A marginal record may correctly state the probability that a project is strong and the probability that its losses reflect growth while omitting whether strength and growth are believed to occur together. Because that association affects the likelihood assigned to future signals, learners with identical marginal records can update differently from a learner who retains the full joint belief. Repeated marginal updating can therefore generate learning dynamics that depart systematically from fully associated Bayesian learning.

The compression studied here is meant to capture a common economic form of simplified inference. Organizations often do not transmit a full posterior distribution or a full case history. They transmit a scorecard, memo, file, dashboard, or standardized evaluation. Experimental evidence suggests that people often underuse dependence information when forming beliefs. Work on correlation neglect shows that subjects may fail to account for correlation or redundancy in signals (Enke and Zimmermann, 2019). The pseudocontingency literature shows that people may infer associations from marginal frequencies rather than from the full joint distribution (Fiedler, 2010; Fiedler and Freytag, 2004). Field evidence on learning through noticing shows that agents may possess relevant data but fail to encode the dimension needed for later learning (Hanna et al., 2014). At the institutional level, high-stakes decisions are increasingly mediated by prediction scores, risk assessments, and algorithmic recommendations (Angelova et al., 2026; Kleinberg et al., 2018; Stevenson and

Doleac, 2024). These papers do not test the exact recursive rule studied here. They motivate the object of interest: a low-dimensional record that preserves separate assessments while omitting dependence information.

We formalize this idea in an environment with a two-component state and an infinite sequence of signals. The true state belongs to the maintained model and can be identified. The learner observes every signal, knows the correct likelihood family, and applies Bayes’ rule whenever she updates a joint belief. Her only restriction concerns the information retained between dates. She stores the two posterior marginals, reconstructs a joint prior by taking their product, applies Bayes’ rule to that reconstructed prior after the next signal, and again retains only the resulting marginals. We call this procedure recursive marginal learning.

The recursive marginal-updating mechanism is closely related to that studied by Loh and Phelan (2019). They show that agents who retain marginal beliefs and reconstruct joint beliefs using independence can interpret common information differently, sustain disagreement, and converge to false limiting beliefs. We take this mechanism as a starting point but pursue a different characterization agenda. We ask which pure marginal records the recursion can select and when the marginal record is sufficient for subsequent Bayesian updating. The first question leads to a two-sided theory of asymptotic selection. The second identifies the exact boundary between lossless and lossy recursive compression.

Our results identify which false states can become self-reinforcing, show that false certainty can arise from ordinary interior beliefs, and characterize exactly when retaining only marginal beliefs is harmless.

Our first result characterizes the states that recursive marginal learning can select. Because the learner repeatedly reconstructs a joint belief from separate marginal beliefs, the learning process does not directly compare every state with every other state. Instead, each marginal update effectively compares states that differ in only one component. A state can therefore become self-reinforcing when it fits the evidence better than every alternative obtained by changing one component at a time, even though another state that differs in both components fits the evidence better overall. Any state reached with positive probability must satisfy the weak version of this condition. When all of the one-component comparisons are strict, the state attracts nearby beliefs, and the probability of escaping its reinforcing region falls exponentially with the strength of the initial beliefs.

A binary example illustrates this result.¹ A project has an unknown type, H or L , and an unknown regime, g or s . The true state is (L, s) , and a binary signal takes the value 1 with

¹This example will be discussed in Section 3.

probabilities

	g	s
H	$3/5$	$1/10$
L	$9/10$	$1/2$.

Full Bayesian learning identifies (L, s) .

Recursive marginal learning can nevertheless become certain of (H, g) . The false state generates signal 1 with probability $3/5$, which is closer to the true probability $1/2$ than either one-component deviation: (L, g) generates 1 with probability $9/10$, while (H, s) generates it with probability $1/10$. Thus, holding the regime fixed at g , the evidence favors H over L ; holding the type fixed at H , it favors g over s . Once the marginal beliefs begin to favor H and g , the two assessments reinforce one another.

The truth defeats the false state only through a joint comparison: (L, s) differs from (H, g) in both components and matches the signal distribution exactly. Full Bayes retains this comparison and learns the truth. Recursive compression discards the association between the two components and therefore loses the information needed to preserve it.

Our second result shows that false learning is not confined to beliefs already concentrated near the false state. In an open class of binary environments, recursive marginal learning converges almost surely, and false certainty occurs with a probability bounded away from zero across broad classes of interior initial beliefs. The result is robust to small changes in the signal distributions and applies even when the initial marginal beliefs are nearly balanced. On the histories leading to false certainty, full Bayes learner knows the truth. The failure is therefore caused by the recursive reuse of the compressed record, not by the initial prior, the statistical model, or marginal reporting itself.

Our third result identifies exactly when recursive compression is universally harmless from product priors. Marginal beliefs reproduce full Bayesian updating from every full-support product prior and along every finite signal history if and only if the likelihood of each signal is multiplicatively separable across the two state components. In that case, Bayesian updating preserves the product form, so the joint belief remains recoverable from its marginals. Recursive marginal learning then coincides pathwise with full Bayes after every signal.

When this condition fails, a signal can change not only the belief about each component but also how the two components are related. The marginals omit this relationship, even though it may be needed to interpret later signals. When both components are binary, the missing information can be summarized by a single additional number measuring their association. The two marginals together with this interaction measure uniquely determine

the full posterior. Retaining this one additional statistic therefore restores exact Bayesian updating.

The broader implication is that marginal assessments are not inherently unreliable. They are guaranteed to be adequate recursive records from product beliefs when new evidence creates no interaction across the recorded dimensions. Otherwise, a record may accurately describe each dimension separately at the moment it is produced while omitting how the dimensions fit together. Reusing that incomplete record can permanently redirect learning, even when the model is correctly specified and every within-period update applies Bayes’ rule correctly.

1.1 Related literature

The closest paper is [Loh and Phelan \(2019\)](#), which studies recursive marginal updating in a broader multidimensional setting and shows that it can generate persistent disagreement and false limiting beliefs. In our finite two-coordinate environment, we characterize pure-state selection through coordinate-wise KL comparisons and establish robust false certainty from interior beliefs. We also show when marginal records reproduce full Bayesian updating.

The paper also contributes to Bayesian learning under misspecification. Classical results show that when the maintained model need not contain the truth, posterior mass concentrates on parameters or states minimizing KL divergence from the true data-generating process ([Berk, 1966](#); [Bunke and Milhaud, 1998](#)). Economic models use the same KL logic to characterize long-run beliefs and behavior under subjective models that may exclude the truth ([Esponda and Pouzo, 2016](#); [Esponda et al., 2021](#)). Our environment differs at the primitive level. The true state belongs to the model, the likelihood family is correct, every signal is observed, and full Bayes is consistent. The learner also applies Bayes’ rule correctly to the joint belief available at each date. The distortion is intertemporal: posterior association created by one update is not retained for the next. Near pure records, this deletion changes the coordinate-wise likelihood comparisons governing subsequent drift.

A related literature studies simplified representations and limited attention. In models of coarse thinking, agents group rich situations into simplified categories and transfer informational content across situations treated as similar ([Gennaioli and Shleifer, 2010](#); [Mullainathan et al., 2008](#)). In models of selective attention, agents may update correctly on attended dimensions while failing to encode variables relevant for later prediction ([Schwartzstein, 2014](#)). Field evidence on learning through noticing similarly shows that agents may possess relevant data but fail to learn from it when they do not encode the predictive dimension ([Hanna et al., 2014](#)). Our learner observes every signal and retains the posterior assessment of each state coordinate. What she discards is only the posterior association between those coordinates.

The closest empirical motivation comes from neglected dependence. Experimental work on correlation neglect shows that subjects may underadjust for correlation or redundancy in information (Enke and Zimmermann, 2019). The pseudocontingency literature shows that people may infer joint relationships from marginal frequencies rather than representing the full contingency structure (Eder et al., 2011; Fiedler, 2010; Fiedler and Freytag, 2004). These findings support the plausibility of low-dimensional representations that omit dependence information. The mechanism here is different: the learner does not misperceive the objective likelihoods or treat contemporaneous signals as independent. Earlier observations create posterior association through correct Bayesian updating, and the record deletes that association before later observations are interpreted.

Other papers study neglected dependence in strategic and social environments. Cursed equilibrium assumes that players underappreciate the dependence between others’ actions and private information (Eyster and Rabin, 2005). Models and experiments in social learning examine whether agents account for redundancy in predecessors’ actions and information (Angrisan et al., 2021; Eyster et al., 2018). Spiegel (2016) represents subjective beliefs through restricted Bayesian networks that omit links among variables. These approaches share the idea that incomplete representations of dependence can distort inference. Our learner instead knows the objective signal technology and does not maintain a fixed subjective causal graph. The omitted dependence is generated endogenously by Bayesian updating and then deleted by the record.

The rest of the paper proceeds as follows. Section 2 defines the environment and distinguishes full Bayesian learning from recursive marginal learning. Section 3 presents the binary false-certainty example. Section 4 gives the two-sided selection theorem. Section 5 establishes the exhaustive and uniform binary result. Section 6 characterizes the rank-one boundary and develops the binary one-scalar repair. Section 7 concludes.

2 Marginal Records and Recursive Learning

The state has two components, and the learner observes an infinite sequence of signals about their joint realization. We study two learning procedures that share the same information and statistical model. Both observe every signal, know the same correctly specified likelihood family, and use the same Bayesian update operator. They differ only in the object retained between dates. A full Bayesian learner retains the joint posterior. A recursive marginal learner retains only the two posterior marginals, reconstructs their product before the next signal, and thereby discards posterior association.

2.1 Environment and learning procedures

For a finite set S , let $\Delta(S)$ denote the set of full-support probability distributions on S , and let $\overline{\Delta}(S)$ denote its closure, the set of all probability distributions on S .² For any finite set E and any $e \in E$, let $\delta_e \in \overline{\Delta}(E)$ denote the degenerate distribution concentrated on e .

Let A , B , and Y be finite sets, with $|A|, |B| \geq 2$, and let the state be $x = (a, b) \in X := A \times B$. Conditional on state (a, b) , the signals $(Y_t)_{t \geq 1}$ are independently and identically distributed according to $P_{ab} \in \Delta(Y)$. The true state is $x^* = (a^*, b^*)$, and the true signal distribution is $P^* := P_{a^*b^*}$. Write $\mathbb{P}^*$ and $\mathbb{E}^*$ for probability and expectation under the true state. Every learner knows the likelihood family $\{P_{ab} : (a, b) \in X\}$, so the statistical model is correctly specified.

For any finite set E and any $\mu, \nu \in \overline{\Delta}(E)$, define

$$D_{\text{KL}}(\mu \| \nu) := \sum_{e \in E} \mu(e) \log \frac{\mu(e)}{\nu(e)},$$

with the convention that terms with $\mu(e) = 0$ are zero and that $D_{\text{KL}}(\mu \| \nu) = +\infty$ if $\nu(e) = 0 < \mu(e)$ for some $e \in E$. The KL distance of state (a, b) from the true signal distribution is

$$K(a, b) := D_{\text{KL}}(P^* \| P_{ab}) = \sum_{y \in Y} P^*(y) \log \frac{P^*(y)}{P_{ab}(y)}. \quad (1)$$

Thus $K(a^*, b^*) = 0$. The true state is called *identified* if $K(a, b) > 0$ for every $(a, b) \neq (a^*, b^*)$.

Let

$$Q := \Delta(A) \times \Delta(B), \quad \overline{Q} := \overline{\Delta}(A) \times \overline{\Delta}(B).$$

A finite-date marginal record is a pair $q = (\alpha, \beta) \in Q$. The closure $\overline{Q}$ also contains the degenerate records used to describe limiting beliefs.

Given a joint belief $\pi \in \overline{\Delta}(X)$, define its marginal record by

$$\mathbf{M}(\pi) := (\pi_A, \pi_B), \quad \pi_A(a) := \sum_{b \in B} \pi(a, b), \quad \pi_B(b) := \sum_{a \in A} \pi(a, b).$$

Given $q = (\alpha, \beta) \in \overline{Q}$, define its product reconstruction $R(q) \in \overline{\Delta}(X)$ by $R(q)(a, b) := \alpha(a)\beta(b)$. A joint belief $\pi \in \overline{\Delta}(X)$ is a *product belief* if $\pi = R(q)$ for some $q \in \overline{Q}$, or

²We reserve the unbarred notation for likelihoods, priors, and finite-date beliefs, all of which have full support in the model. Full support keeps likelihood ratios finite and is preserved by Bayesian updating. The closure $\overline{\Delta}(S)$ is used only for degenerate limiting beliefs and for results extended to the boundary.

equivalently if

$$\pi(a, b) = \pi_A(a)\pi_B(b) \quad \text{for every } a \in A, b \in B.$$

Thus

$$\mathbf{M}(R(q)) = q, \quad R(\mathbf{M}(\pi))(a, b) = \pi_A(a)\pi_B(b). \quad (2)$$

The operator $R \circ \mathbf{M}$ preserves the two marginals of a joint belief and removes their association.

Given $\pi \in \overline{\Delta}(X)$ and $y \in Y$, define the Bayesian update operator

$$\mathbf{B}_y(\pi)(a, b) := \frac{\pi(a, b)P_{ab}(y)}{\sum_{a' \in A} \sum_{b' \in B} \pi(a', b')P_{a'b'}(y)}. \quad (3)$$

Because every likelihood has full support, this operator is well defined on $\overline{\Delta}(X)$ and maps $\Delta(X)$ into itself. Define the one-period recursive update by

$$U_y(q) := \mathbf{M}(\mathbf{B}_y(R(q))), \quad q \in \overline{Q}. \quad (4)$$

The map U_y sends Q into Q . The two learning procedures differ not in the Bayesian update operator, but in the object retained after the update is completed.

Full Bayesian learning. A full Bayesian learner retains a joint posterior $\pi_t^F \in \Delta(X)$. Starting from $\pi_0^F \in \Delta(X)$, she updates according to

$$\pi_{t+1}^F := \mathbf{B}_{Y_{t+1}}(\pi_t^F). \quad (5)$$

For any $(a, b), (a', b') \in X$, iterating (5) gives

$$\log \frac{\pi_t^F(a, b)}{\pi_t^F(a', b')} = \log \frac{\pi_0^F(a, b)}{\pi_0^F(a', b')} + \sum_{\tau=1}^t \log \frac{P_{ab}(Y_\tau)}{P_{a'b'}(Y_\tau)}. \quad (6)$$

The full Bayesian learner therefore preserves every posterior comparison among joint states.

Recursive marginal learning. Starting from $q_0 \in Q$, the recursive marginal learner retains the marginal record $q_t = (\alpha_t, \beta_t) \in Q$, which evolves according to

$$q_{t+1} := U_{Y_{t+1}}(q_t) = \mathbf{M}(\mathbf{B}_{Y_{t+1}}(R(q_t))). \quad (7)$$

At date $t+1$, she reconstructs the product prior $R(q_t)$, forms the correct one-period posterior $\mathbf{B}_{Y_{t+1}}(R(q_t))$, and stores its two marginals as q_{t+1} . The joint posterior formed within the date is not retained.

The learner therefore marginalizes correctly after every one-period Bayesian update. Her restriction is entirely intertemporal: posterior association generated at one date cannot be carried into the interpretation of later signals. Such a restriction can arise when decision makers are short-lived, when responsibility passes across agents through standardized summaries, or when an information system stores separate assessments for multiple dimensions without retaining their joint distribution.³

Whenever the two procedures are compared, fix $q_0 \in Q$ and initialize the full Bayesian learner from the same product belief, $\pi_0^F := R(q_0)$. The dynamic state variables are π_t^F , the full-history joint posterior generated by (5), and q_t , the recursively reused marginal record generated by (7). The object $M(\pi_t^F)$ is only the marginal report of the full Bayesian posterior at date t . It is formed after the history through date t has been processed and is not used to process the next signal.

Although $q_0 = M(\pi_0^F)$, the recursive record q_t need not equal $M(\pi_t^F)$ at later dates. The recursive learner repeatedly feeds q_t back into future updating through $R(q_t)$, whereas the full Bayesian learner retains the association contained in π_t^F .

2.2 Static projection and the Bayesian benchmark

The product reconstruction used by the recursive marginal learner is not an arbitrary approximation. At any fixed date, the product of the two marginals is the forward-KL-optimal product representation of the joint belief.

Proposition 1. *For every $\pi \in \Delta(X)$, the marginal record $M(\pi)$ is the unique solution to*

$$\min_{q \in \overline{Q}} D_{\text{KL}}(\pi \| R(q)).$$

Proposition 1 establishes the static quality of the representation. Given any joint belief, the learner stores exactly the marginal record whose product reconstruction is closest to that belief in forward KL divergence. The substantive question is therefore not whether the learner chooses the wrong product representation at a given date. It is whether that representation remains sufficient when it is repeatedly reused to interpret subsequent signals.

The next proposition records the unrestricted Bayesian benchmark.

³Examples include organizations with frequent personnel turnover, committees whose members inherit only attribute-specific assessments from their predecessors, and information systems that maintain separate scores or dashboards for multiple dimensions without retaining their full joint distribution. In each case, the current decision maker may update correctly using the available marginal record while lacking the association generated by earlier observations.

Proposition 2. *Let $(\pi_t^F)_{t \geq 0}$ be the full Bayesian posterior process generated from any $\pi_0^F \in \Delta(X)$. If the true state is identified, then*

$$\pi_t^F \rightarrow \delta_{(a^*, b^*)}, \quad \mathbf{M}(\pi_t^F) \rightarrow (\delta_{a^*}, \delta_{b^*}) \quad \mathbb{P}^* \text{-a.s.}$$

The first conclusion is standard Bayesian consistency. The second result shows that marginal reporting is asymptotically harmless when it is applied to the full-history posterior after the available signal history has already been processed.

When the full Bayesian learner and the recursive learner are initialized from the common product belief $\pi_0^F = R(q_0)$, any divergence between q_t and $\mathbf{M}(\pi_t^F)$ must arise from the recursive reuse of the compressed record in (7). It cannot be attributed to misspecified likelihoods, discarded signals, incorrect within-period Bayesian updating, or a statically inferior product representation. The remainder of the paper studies the dynamic consequences of this repeated deletion of posterior association.

3 A Binary False-Certainty Example

The results above show that any distortion under recursive marginal learning must come from the repeated deletion and reuse of posterior association, rather than from the static quality of the product representation. This section presents a minimal binary environment in which the two stored marginals reinforce one another and draw the recursive marginal learner toward a false state. The example previews the coordinate-wise KL criterion developed in the later section.

Let

$$A = \{H, L\}, \quad B = \{g, s\}, \quad Y = \{0, 1\}.$$

The first coordinate describes project type, where H denotes a strong project and L a weak project. The second coordinate describes the diagnostic regime, where g denotes a growth regime and s a solvency-risk regime. The realization $Y = 1$ is diagnostic and need not represent project success. Conditional on state (a, b) , the signal is Bernoulli, with

$P_{ab}(1)$	g	s	and	$P_{ab}(0) = 1 - P_{ab}(1).$
H	$3/5$	$1/10$		
L	$9/10$	$1/2$		

The true state is

$$x^* = (L, s), \quad P^* = P_{Ls} = \text{Bernoulli}(1/2).$$

The four signal distributions are distinct, so the true state is identified. Proposition 2 therefore implies that, from any full-support prior,

$$\pi_t^F \rightarrow \delta_{(L,s)}, \quad \mathbf{M}(\pi_t^F) \rightarrow (\delta_L, \delta_s) \quad \mathbb{P}^*\text{-a.s.}$$

Using the KL distance defined in (1), the distances from the true signal distribution are

$$K(L, s) = 0, \quad K(H, g) = \frac{1}{2} \log \frac{25}{24}, \quad K(H, s) = K(L, g) = \frac{1}{2} \log \frac{25}{9}. \quad (8)$$

Thus (L, s) is the unique global KL minimizer. Nevertheless, the false state $\bar{x} = (H, g)$ fits the true signal distribution better than either state obtained by changing only one of its coordinates:

$$K(H, g) < K(L, g), \quad K(H, g) < K(H, s). \quad (9)$$

The false state is therefore locally best along both coordinate slices through (H, g) , even though the diagonal alternative (L, s) fits the true signal distribution perfectly.

To see how the local advantage in (9) affects the recursive dynamics in (7), write $q_t = (\alpha_t, \beta_t)$. The type update compares H with L using likelihoods averaged according to the current regime record. In particular, it compares

$$\beta_t(g)P_{Hg}(y) + \beta_t(s)P_{Hs}(y) \quad \text{with} \quad \beta_t(g)P_{Lg}(y) + \beta_t(s)P_{Ls}(y).$$

As $\beta_t(g)$ approaches one, the conditional expected change in the log odds of H against L approaches the expected log-likelihood ratio between P_{Hg} and P_{Lg} . Similarly, the regime update compares

$$\alpha_t(H)P_{Hg}(y) + \alpha_t(L)P_{Lg}(y) \quad \text{with} \quad \alpha_t(H)P_{Hs}(y) + \alpha_t(L)P_{Ls}(y),$$

so, as $\alpha_t(H)$ approaches one, the conditional expected change in the log odds of g against s approaches the expected log-likelihood ratio between P_{Hg} and P_{Hs} .

At the pure record (δ_H, δ_g) , these limiting expected changes are

$$\begin{aligned} \sum_{y \in Y} P^*(y) \log \frac{P_{Hg}(y)}{P_{Lg}(y)} &= K(L, g) - K(H, g) = \frac{1}{2} \log \frac{8}{3} > 0, \\ \sum_{y \in Y} P^*(y) \log \frac{P_{Hg}(y)}{P_{Hs}(y)} &= K(H, s) - K(H, g) = \frac{1}{2} \log \frac{8}{3} > 0. \end{aligned} \tag{10}$$

Because these expected log-odds changes depend continuously on the current marginal record, the positive drifts in (10) remain strictly positive whenever q_t is sufficiently close to (δ_H, δ_g) .

The reinforcement operates across coordinates. A high stored probability on g makes the type update favor H , because P_{Hg} fits the data better than P_{Lg} . A high stored probability on H makes the regime update favor g , because P_{Hg} fits the data better than P_{Hs} . Each stored marginal therefore selects the likelihood comparison that reinforces the other.

The full Bayesian learner retains a direct posterior comparison between the true diagonal state (L, s) and the false diagonal state (H, g) . Applying the posterior log-odds identity (6) and using (8) gives

$$\frac{1}{t} \log \frac{\pi_t^F(L, s)}{\pi_t^F(H, g)} \rightarrow K(H, g) - K(L, s) = \frac{1}{2} \log \frac{25}{24} > 0 \quad \mathbb{P}^*\text{-a.s.} \tag{11}$$

The same signal history therefore drives the full Bayesian posterior toward the truth, and its ex post marginal report $\mathbf{M}(\pi_t^F)$ converges to (δ_L, δ_s) .

Recursive marginal learning does not preserve the diagonal posterior comparison in (11). Each coordinate is updated using the current stored marginal of the other coordinate. Near (H, g) , changing the type alone appears unfavorable, and changing the regime alone also appears unfavorable. Reaching the truth requires revising both coordinates together, but the recursively stored record does not retain the association needed to sustain that joint comparison.

The example reveals the central geometry of the paper. Full Bayesian learning ranks joint states according to their global KL fit. Recursive marginal learning instead generates coordinate-wise comparisons because each coordinate is updated against the stored marginal of the other. A false state can consequently become locally self-reinforcing when it outperforms every one-coordinate deviation from itself, even though the truth is uniquely optimal under global KL distance. The next section formalizes this coordinate-wise geometry and shows when such local reinforcement produces attraction to a pure marginal record.

4 Local Coordinate-wise KL Geometry

The binary example shows that recursive marginal learning need not compare joint states according to their global KL fit. Under product reconstruction, each coordinate is updated using the same stored marginal distribution over the other coordinate. The resulting likelihood comparisons are therefore local to coordinate slices of the state space. This section derives that geometry for arbitrary finite A , B , and Y . Strict coordinate-wise KL minima generate local attraction, while any pure marginal record reached with positive probability must satisfy the corresponding weak coordinate-wise inequalities.

Throughout this section, let $q_t = (\alpha_t, \beta_t)$ denote the recursive marginal record generated from $q_0 \in Q$ through (7). All probabilities are taken under the true signal law $\mathbb{P}^*$.

4.1 Coordinate-wise fit

Fix a candidate state $\bar{x} = (\bar{a}, \bar{b})$. Near this pure marginal record, the recursive marginal learner compares $\bar{a}$ with alternative A -coordinates while holding the B -coordinate close to $\bar{b}$, and compares $\bar{b}$ with alternative B -coordinates while holding the A -coordinate close to $\bar{a}$. A state that differs from $\bar{x}$ in both coordinates is not directly involved in either local marginal comparison. This observation motivates the following definition.

Definition 1 (Coordinate-wise KL minimum). A state $\bar{x} = (\bar{a}, \bar{b})$ is a *weak coordinate-wise KL minimum* if

$$K(\bar{a}, \bar{b}) \leq K(a, \bar{b}) \quad \text{for every } a \neq \bar{a}, \quad K(\bar{a}, \bar{b}) \leq K(\bar{a}, b) \quad \text{for every } b \neq \bar{b}. \quad (12)$$

It is a *strict coordinate-wise KL minimum* if all the inequalities in (12) are strict.

Coordinate-wise KL minimality is weaker than global KL optimality. A state may fit the evidence better than every alternative obtained by changing one coordinate while fitting the evidence worse than a state that differs in both coordinates. Full Bayesian learning preserves the latter comparison. Recursive marginal learning may not, because each coordinate is revised using the current stored assessment of the other.

Under identification, the true state $x^* = (a^*, b^*)$ is a strict coordinate-wise KL minimum: it has KL distance zero, whereas every false state has strictly positive KL distance.⁴

⁴Under identification, any false weak coordinate-wise KL minimum must differ from the true state in both coordinates. If it shared either coordinate with the truth, the true state would be one of its one-coordinate deviations and would fit the data strictly better.

4.2 Selection and local attraction

The next theorem provides complementary necessary and sufficient conditions for pure-state selection and local attraction.

Theorem 1. Fix $\bar{x} = (\bar{a}, \bar{b})$.

- (i) Necessity. For any $q_0 \in Q$, if $\mathbb{P}^*(q_t \rightarrow (\delta_{\bar{a}}, \delta_{\bar{b}})) > 0$, then $\bar{x}$ is a weak coordinate-wise KL minimum.
- (ii) Local sufficiency. If $\bar{x}$ is a strict coordinate-wise KL minimum, then, for every $\varepsilon \in (0, 1)$, there exists $\eta_\varepsilon \in (0, 1)$ such that every initial record $q_0 = (\alpha_0, \beta_0) \in Q$ satisfying

$$\alpha_0(\bar{a}) \geq \eta_\varepsilon, \quad \beta_0(\bar{b}) \geq \eta_\varepsilon$$

also satisfies

$$\mathbb{P}^*(q_t \rightarrow (\delta_{\bar{a}}, \delta_{\bar{b}})) \geq 1 - \varepsilon.$$

The theorem leaves a gap only when the candidate is tied in KL distance with one of its one-coordinate deviations. When such ties are absent, weak and strict coordinate-wise KL minimality coincide, yielding the following exact characterization.

Corollary 1. Fix $\bar{x} = (\bar{a}, \bar{b})$, and suppose

$$K(\bar{a}, \bar{b}) \neq K(a, \bar{b}) \quad \text{for every } a \neq \bar{a}, \quad K(\bar{a}, \bar{b}) \neq K(\bar{a}, b) \quad \text{for every } b \neq \bar{b}.$$

Then the following statements are equivalent.

- (i) The state $\bar{x}$ is a strict coordinate-wise KL minimum.
- (ii) For every $\varepsilon \in (0, 1)$, there exists $\eta_\varepsilon \in (0, 1)$ such that every initial record $q_0 = (\alpha_0, \beta_0) \in Q$ satisfying

$$\alpha_0(\bar{a}) \geq \eta_\varepsilon, \quad \beta_0(\bar{b}) \geq \eta_\varepsilon$$

also satisfies

$$\mathbb{P}^*(q_t \rightarrow (\delta_{\bar{a}}, \delta_{\bar{b}})) \geq 1 - \varepsilon.$$

- (iii) There exists a full-support initial record $q_0 \in Q$ such that $\mathbb{P}^*(q_t \rightarrow (\delta_{\bar{a}}, \delta_{\bar{b}})) > 0$.

Theorem 1 provides the two directions behind this characterization. Part (i) shows that a pure state can be selected with positive probability only if it satisfies the weak inequalities in (12): no change in either coordinate, holding the other fixed, improves the fit with the observed data.

Part (ii) gives the local converse. If every one-coordinate deviation fits the data strictly worse, sufficiently strong beliefs in $\bar{a}$ and $\bar{b}$ reinforce one another. A high stored probability on $\bar{b}$ makes the evidence favor $\bar{a}$, while a high stored probability on $\bar{a}$ makes the evidence favor $\bar{b}$. Once both marginal beliefs are sufficiently concentrated, the probability of converging to the associated pure record can be made arbitrarily close to one.

The appendix establishes a stronger quantitative version of part (ii). Once the record lies in a reinforcing region, the probability of escaping that region decreases exponentially with the smallest initial log-odds advantage assigned to the candidate coordinates. The theorem is nevertheless local: it does not assert that the recursive record converges from every initial condition.⁵

Corollary 1 converts these two directions into an exact characterization when the candidate has no coordinate-wise KL ties. Its initial-condition quantifier is existential: the corollary identifies whether the pure state can be selected from some full-support initial record, not whether it is selected from every fixed initial record.

The economic mechanism is a local belief trap. The learner evaluates each coordinate while treating the current assessment of the other coordinate as the relevant anchor. She may therefore fail to recognize that both coordinates should be revised together. A state need not be the best overall explanation of the data; it can attract recursive learning whenever neither coordinate appears worth changing on its own.

If $\bar{x} \neq x^*$ is a strict coordinate-wise KL minimum, Theorem 1 implies that recursive marginal learning can become falsely certain of $\bar{x}$. On the corresponding convergence event,

$$R(q_t)(\bar{a}, \bar{b}) = \alpha_t(\bar{a})\beta_t(\bar{b}) \rightarrow 1.$$

Under identification, the full Bayesian posterior nevertheless converges to the true state. The difference arises because full Bayesian learning retains comparisons between states that differ in both coordinates, whereas recursive marginal learning reduces future inference to separate one-coordinate comparisons.

In the binary example of Section 3, the inequalities in (9) show that the false state (H, g) has exactly this property. It fits the evidence better than (L, g) and (H, s) , although the true diagonal state (L, s) fits the evidence best overall. The next section shows that the reinforcing region around (H, g) can be reached with positive probability from broad classes of interior initial records and that this false-learning result survives small perturbations of the likelihood system.

⁵In the appendix B, we provide a sufficient condition for almost-sure point convergence.

5 Robust Same-History False Learning

The preceding section establishes local attraction: a strict coordinate-wise KL minimum attracts records that already assign sufficiently high probability to its two coordinates. We now strengthen that result in the binary environment of Section 3. The reinforcing region around the false state (H, g) can be reached from every full-support initial record after a suitable finite signal history. Moreover, this accessibility and the resulting false-learning probability survive perturbations of the likelihood system.

The result provides a same-history comparison between full Bayesian learning and recursive marginal learning. Both learners begin from the same product belief, observe the same signals, and know the same likelihood system. The full Bayesian learner learns the true state almost surely, whereas the recursive learner becomes certain of the false state with positive probability.

Let

$$\mathcal{P} := \prod_{(a,b) \in X} \Delta(Y)$$

denote the space of full-support likelihood systems. We equip $\mathcal{P}$ with the relative Euclidean topology inherited from $\prod_{(a,b) \in X} \mathbb{R}^{|Y|}$. Because $\Delta(Y)$ consists of full-support distributions, every $P \in \mathcal{P}$ has strictly positive likelihoods.

Let $P^0 = (P_{ab}^0)_{(a,b) \in X} \in \mathcal{P}$ denote the baseline likelihood system from Section 3. Consider a likelihood system $P = (P_{ab})_{(a,b) \in X} \in \mathcal{P}$ near P^0 . For every such P , retain $x^* = (L, s)$ as the true state and let signals be independently generated according to P_{Ls} . Write $\mathbb{P}^P$ for the resulting probability law.

Fix $q_0 \in Q$. Initialize the full Bayesian learner at $\pi_0^F = R(q_0)$, and initialize the recursive marginal learner at q_0 . The processes π_t^F and q_t are therefore generated from the same realized signal history.

For $\zeta > 0$, define

$$Q_\zeta := \left\{ (\alpha, \beta) \in Q : \left| \log \frac{\alpha(H)}{\alpha(L)} \right| \leq \zeta, \left| \log \frac{\beta(g)}{\beta(s)} \right| \leq \zeta \right\}. \quad (13)$$

The class Q_ζ in (13) contains initial records whose marginal odds are not arbitrarily extreme. Every full-support marginal record belongs to Q_ζ for some finite ζ .

Theorem 2. *There exists an open neighborhood $\mathcal{O}$ of P^0 such that the following statements hold.*

(i) For every $P \in \mathcal{O}$ and every $q_0 \in Q$, the full Bayesian learner learns the true state:

$$\pi_t^F \rightarrow \delta_{(L,s)} \quad \mathbb{P}^P\text{-a.s.}$$

(ii) For every $P \in \mathcal{O}$ and every $q_0 \in Q$, the recursive marginal record converges almost surely to one of the two diagonal pure records:

$$\mathbb{P}^P (q_t \rightarrow (\delta_L, \delta_s) \text{ or } q_t \rightarrow (\delta_H, \delta_g)) = 1.$$

Both limiting events have strictly positive probability. Moreover, for every $\zeta > 0$, there exists $\eta_\zeta > 0$ such that

$$\inf_{P \in \mathcal{O}} \inf_{q_0 \in Q_\zeta} \mathbb{P}^P (q_t \rightarrow (\delta_H, \delta_g)) \geq \eta_\zeta.$$

Part (i) shows that small perturbations of the likelihoods do not compromise identification. The full Bayesian posterior continues to converge to (L, s) . Consequently, its ex post marginal report also satisfies

$$\mathbf{M}(\pi_t^F) \rightarrow (\delta_L, \delta_s) \quad \mathbb{P}^P\text{-a.s.}$$

This marginal report is formed from the full-history posterior and is not reused in subsequent updating.

Part (ii) shows that recursive marginal learning has two possible long-run outcomes throughout the same neighborhood. Both the true state (L, s) and the false state (H, g) remain strict coordinate-wise KL minima, and both associated pure records have strictly positive probability from every full-support initial record. The recursive record converges almost surely to one of them. In addition, the false limit retains a probability bounded away from zero whenever the initial marginal odds are uniformly bounded as in (13).

The result is not restricted to nearly balanced initial records. Every $q_0 \in Q$ belongs to Q_ζ for some finite ζ , so the theorem implies

$$\mathbb{P}^P (q_t \rightarrow (\delta_H, \delta_g)) > 0$$

for every $P \in \mathcal{O}$ and every full-support initial record q_0 . The constant η_ζ may decrease as ζ increases. Larger values of ζ allow initial marginal records closer to the boundary of Q , including records that strongly favor L or s . Moving such records into the reinforcing region around (H, g) may require a longer and less probable signal history. The theorem

therefore provides a uniform lower bound for each bounded log-odds class, but does not claim a common lower bound over all interior initial records.

The separation occurs along the same realized histories, not merely across different probability distributions. For each fixed P and q_0 , both learning procedures are determined by the same signal sequence. Because full Bayesian consistency holds almost surely, there is a positive-probability set of histories on which

$$\pi_t^F \rightarrow \delta_{(L,s)}, \quad \mathbf{M}(\pi_t^F) \rightarrow (\delta_L, \delta_s), \quad q_t \rightarrow (\delta_H, \delta_g). \quad (14)$$

The different limits in (14) therefore arise solely from the recursive treatment of posterior association. Full Bayesian learning retains the association generated by earlier signals. Its marginal report is taken only after the available history has been processed. Recursive marginal learning instead deletes that association before each subsequent signal is interpreted.

The next section identifies the boundary at which this same-history divergence is impossible. It characterizes the likelihood systems for which a marginal record contains everything needed for future Bayesian updating.

6 When Marginal Records Are Dynamically Sufficient

The preceding sections show that recursive marginal learning can diverge from full Bayesian learning because posterior association is deleted before later signals are interpreted. This section identifies the boundary at which that deletion is harmless. Marginal records are dynamically sufficient when Bayesian updating from an independent prior does not create association between the two state coordinates.

We first characterize the likelihood systems satisfying this requirement. Under the resulting no-interaction condition, recursive marginal learning reproduces full Bayesian learning from every product prior and along every finite signal history. We then specialize to a 2×2 state space. In that case, the marginal record omits exactly one dimension of the joint posterior, and storing one additional measure of association restores exact Bayesian updating.

6.1 The no-interaction condition

We say that the likelihood system has *no coordinate interaction* if, for every signal $y \in Y$, there exist strictly positive functions f_A^y and f_B^y such that

$$P_{ab}(y) = f_A^y(a)f_B^y(b) \quad \text{for every } a \in A, b \in B. \quad (15)$$

Under (15), the likelihood contribution of a signal separates across the two state coordinates. Starting from an independent belief, the signal changes the two marginal beliefs without creating association between them.

Theorem 3. *The following statements are equivalent.*

- (i) *The likelihood system has no coordinate interaction.*
- (ii) *For every $q \in Q$ and every $y \in Y$, the posterior $B_y(R(q))$ is a product belief.*
- (iii) *For every $q_0 \in Q$ and every finite signal history, if the full Bayesian learner is initialized at $\pi_0^F = R(q_0)$, then*

$$R(q_t) = \pi_t^F$$

at every date along that history.

Theorem 3 characterizes universal dynamic sufficiency of marginal records from product priors. Under the no-interaction condition in (15), every Bayesian update preserves independence. Marginalization therefore removes no information, and the recursive learner's product reconstruction is exactly the full Bayesian posterior after every signal.

Conversely, suppose (15) fails for some signal. That signal creates posterior association from at least one full-support product prior. The resulting posterior and its product reconstruction have the same marginals but are different joint beliefs. Because the two beliefs can assign different likelihoods to subsequent signals, the marginal record is no longer guaranteed to reproduce full Bayesian updating. The two procedures may still coincide for particular priors or histories, but not universally.

The no-interaction condition also makes KL fit additively separable across coordinates. Under the factorization in (15), there exist a constant c and functions κ_A and κ_B such that⁶

$$K(a, b) = c + \kappa_A(a) + \kappa_B(b), \quad (16)$$

⁶We formally establish this decomposition in the proof of Theorem 3.

where

$$\begin{aligned} c &:= \sum_{y \in Y} P^*(y) \log P^*(y), \\ \kappa_A(a) &:= - \sum_{y \in Y} P^*(y) \log f_A^y(a), \\ \kappa_B(b) &:= - \sum_{y \in Y} P^*(y) \log f_B^y(b). \end{aligned}$$

A state is therefore a weak coordinate-wise KL minimum if and only if each of its coordinates minimizes the corresponding component of the additive decomposition in (16). It must consequently be a global KL minimum. Under identification, the true state is the unique global KL minimum and hence the unique weak coordinate-wise KL minimum. In particular, no false strict coordinate-wise KL minimum can exist.

6.2 The binary one-scalar repair

We now specialize to a 2×2 state space. The result applies to every full-support binary likelihood system, not only to the example in Section 3. Let

$$A = \{a_0, a_1\}, \quad B = \{b_0, b_1\}.$$

A joint belief over four states has three independent probabilities,⁷ whereas its two binary marginals contain only two: the marginal record records only the probability of a_0 and the probability of b_0 . The marginal record therefore omits exactly one degree of freedom, namely, the association between the two coordinates.

For any $\pi \in \Delta(X)$, define its interaction coordinate by

$$\ell(\pi) := \log \frac{\pi(a_1, b_1) \pi(a_0, b_0)}{\pi(a_1, b_0) \pi(a_0, b_1)}. \quad (17)$$

The log cross-ratio in (17) compares the relative posterior weight placed on the matched pairs (a_1, b_1) and (a_0, b_0) with the weight placed on the crossed pairs (a_1, b_0) and (a_0, b_1) . Equivalently, it measures how the odds between a_1 and a_0 vary with the realization of B .⁸ In

⁷For example, if we specify the probabilities of (a_0, b_0) , (a_0, b_1) , and (a_1, b_0) , the remaining probability is determined.

⁸Indeed,

$$\exp(\ell(\pi)) = \frac{\pi(a_1 | b_1) / \pi(a_0 | b_1)}{\pi(a_1 | b_0) / \pi(a_0 | b_0)}.$$

Thus, $\ell(\pi) = 0$ exactly when the realization of B does not change the odds between a_1 and a_0 , which is equivalent to independence in a full-support 2×2 distribution. Full support ensures that all probabilities and odds in this expression are strictly positive and well defined.

particular, $\ell(\pi) = 0$ when the two coordinates are independent, $\ell(\pi) > 0$ when the matched pairs receive relatively greater weight, and $\ell(\pi) < 0$ when the crossed pairs receive relatively greater weight.

For each signal $y \in Y$, define its likelihood interaction by

$$\lambda_y := \log \frac{P_{a_1 b_1}(y) P_{a_0 b_0}(y)}{P_{a_1 b_0}(y) P_{a_0 b_1}(y)}. \quad (18)$$

Corollary 2. *In a full-support 2×2 environment, the pair $(\mathbf{M}(\pi), \ell(\pi))$ uniquely determines every joint belief $\pi \in \Delta(X)$. Moreover, for every $y \in Y$,*

$$\ell(\mathbf{B}_y(\pi)) = \ell(\pi) + \lambda_y, \quad \ell(R(\mathbf{M}(\pi))) = 0. \quad (19)$$

Consequently, if the full Bayesian learner begins from a product prior $\pi_0^F = R(q_0)$, then

$$\ell(\pi_t^F) = \sum_{\tau=1}^t \lambda_{Y_\tau}, \quad (20)$$

and the pair $(\mathbf{M}(\pi_t^F), \ell(\pi_t^F))$ uniquely determines the full Bayesian posterior at every date.

The corollary gives an exact repair of the marginal record. In a binary environment, a learner can retain the two current marginals together with the interaction coordinate in (17). These three numbers uniquely identify the joint posterior, so the learner can reconstruct that posterior before processing the next signal and then store the updated marginals and interaction. No additional information from the earlier signal history is required.

The unaugmented marginal record is universally sufficient if and only if $\lambda_y = 0$ for every signal y . By (18), this is exactly the no-interaction condition of Theorem 3. When every likelihood interaction is zero, a product prior remains product after every signal, and there is no association for compression to delete. When $\lambda_y \neq 0$ for some signal, that signal creates association that cannot be recovered from the two posterior marginals alone.

Recursive marginal learning does not carry this association forward. After each update, the learner retains the two marginals and reconstructs their product before the next signal, thereby resetting the interaction coordinate to zero, as shown in (19). Full Bayesian learning instead preserves the existing interaction, and the contributions of successive signals accumulate according to (20).

In the binary example, the interaction coordinate measures the posterior weight assigned to the diagonal states (H, g) and (L, s) relative to the crossed states (H, s) and (L, g) . This

is precisely the association that matters in the example: the false coordinate-wise attractor (H, g) and the true state (L, s) lie on the same diagonal. Recursive learning preserves the separate probabilities assigned to H and g , but it discards whether those assessments are associated with one another. The one-scalar repair identifies this omitted association as the missing state variable.

7 Conclusion

This paper studies a Bayesian learner who repeatedly replaces a joint posterior with the product of its marginals. The learner observes every signal, knows the correct likelihood family, and applies Bayes' rule within each period. The distortion arises only between periods: posterior association created by one update is deleted before the next signal is interpreted.

Near a pure record, the drift of each marginal log odds is governed by KL comparisons with one-coordinate deviations. Every positive-probability pure limit must therefore be a weak coordinate-wise KL minimum, while every strict coordinate-wise KL minimum attracts sufficiently concentrated records. Away from coordinate-wise ties, they characterize whether a pure state can be selected with positive probability from some full-support initial record. The binary analysis shows that the local geometry can determine global behavior: throughout an identified open neighborhood, the recursive record converges almost surely from every full-support initial condition to one of two diagonal vertices. For each bounded log-odds class, false convergence is bounded away from zero uniformly over the neighborhood and the initial records in that class, even though the full-history posterior learns the truth along the same histories.

Marginal records are universally sufficient for dynamic updating from product priors if and only if the likelihood of every signal is multiplicatively separable across the two state coordinates. Under this condition, updating preserves the product form and creates no association for compression to delete. Outside it, correct marginal reports may omit information needed for later inference. In a binary state space, the posterior log cross-ratio is the missing state variable: storing it with the two marginals restores exact Bayesian updating. The broader implication is that a record should be evaluated not only by whether it accurately reports current marginal assessments, but also by whether it retains the association required to interpret future evidence.

A Proofs

Throughout Appendix A, recall that $\Delta(S)$ and Q contain full-support distributions, while $\overline{\Delta}(S)$ and $\overline{Q}$ also include boundary distributions. Let $\mathcal{F}_t := \sigma(Y_1, \dots, Y_t)$ denote the natural filtration. Unless a likelihood system is displayed explicitly, probabilities, expectations, and almost-sure statements are taken under $\mathbb{P}^*$.

When the likelihood system varies, recall that $\mathcal{P} = \prod_{(a,b) \in X} \Delta(Y)$ denotes the likelihood space equipped with the relative Euclidean topology defined in Section 5. Fixing the true state $x^* = (a^*, b^*)$, for every $P \in \mathcal{P}$ define

$$K^P(a, b) := D_{\text{KL}}(P_{a^*b^*} \| P_{ab}), \quad \mathbb{B}_y^P(\pi)(a, b) := \frac{\pi(a, b)P_{ab}(y)}{\sum_{a' \in A} \sum_{b' \in B} \pi(a', b')P_{a'b'}(y)}.$$

Define $U_y^P(q) := \mathbb{M}(\mathbb{B}_y^P(R(q)))$. Let $\mathbb{P}^P$ and $\mathbb{E}^P$ denote probability and expectation when signals are independently generated according to $P_{a^*b^*}$. For $q = (\alpha, \beta) \in \overline{Q}$, define the effective likelihoods

$$\tilde{P}_A^P(y | a; \beta) := \sum_{b \in B} \beta(b)P_{ab}(y), \quad \tilde{P}_B^P(y | b; \alpha) := \sum_{a \in A} \alpha(a)P_{ab}(y). \quad (21)$$

Whenever P is fixed, its superscript is omitted. The notation $\mathbb{P}_q^P$ is used when a recursively generated event is viewed as a function of the deterministic initial record q ; the underlying signal law remains $\mathbb{P}^P$.

For a candidate state $\bar{x} = (\bar{a}, \bar{b})$ and $C > 0$, define the technical reinforcing region

$$\mathcal{G}_C(\bar{x}) := \left\{ (\alpha, \beta) \in Q : \min_{a \neq \bar{a}} \log \frac{\alpha(\bar{a})}{\alpha(a)} \geq C, \min_{b \neq \bar{b}} \log \frac{\beta(\bar{b})}{\beta(b)} \geq C \right\}. \quad (22)$$

For a finite signal word $w = (y_1, \dots, y_T)$, let $U_w^P := U_{y_T}^P \circ \dots \circ U_{y_1}^P$, with the empty word inducing the identity map. For $x = (a, b)$ and $q = (\alpha, \beta) \in \overline{Q}$, define its product mass by

$$m_x(q) := \alpha(a)\beta(b). \quad (23)$$

For fixed P , say that x is *finitely accessible away from zero product mass* if, for every $\varepsilon \in (0, 1)$ and every $C > 0$, there is a finite signal word w such that $U_w^P(q) \in \mathcal{G}_C(x)$ for every $q \in Q$ satisfying $m_x(q) \geq \varepsilon$.

For a subset S of a finite-dimensional simplex or likelihood space, $\overline{S}$ denotes its closure in the corresponding relative Euclidean topology.

A.1 Proofs of the Main-Text Results

Proof of Proposition 1

Fix $\pi \in \Delta(X)$ and $q = (\alpha, \beta) \in \overline{Q}$. If $\alpha(a) = 0$ for some a or $\beta(b) = 0$ for some b , then $R(q)$ assigns zero probability to a state to which π assigns positive probability, so $D_{\text{KL}}(\pi \| R(q)) = +\infty$. It therefore suffices to consider $q \in Q$. For such q ,

$$D_{\text{KL}}(\pi \| R(q)) = D_{\text{KL}}(\pi \| R(\mathbf{M}(\pi))) + D_{\text{KL}}(\pi_A \| \alpha) + D_{\text{KL}}(\pi_B \| \beta). \quad (24)$$

Indeed, (24) follows by writing

$$\log \frac{\pi(a, b)}{\alpha(a)\beta(b)} = \log \frac{\pi(a, b)}{\pi_A(a)\pi_B(b)} + \log \frac{\pi_A(a)}{\alpha(a)} + \log \frac{\pi_B(b)}{\beta(b)}$$

and summing against $\pi(a, b)$. The first term on the right-hand side of (24) is independent of q , while the last two terms are nonnegative and vanish simultaneously if and only if $\alpha = \pi_A$ and $\beta = \pi_B$. Hence $\mathbf{M}(\pi)$ is the unique minimizer. $\blacksquare$

Proof of Proposition 2

Fix a false state $x = (a, b) \neq x^*$, and write $P_x := P_{ab}$. Because the initial belief $\pi_0^F \in \Delta(X)$ has full support, both $\pi_0^F(x)$ and $\pi_0^F(x^*)$ are strictly positive. Hence the posterior odds between x and x^* are well defined at every finite date.

Applying the full Bayesian log-odds identity (6) to x and x^* , and dividing by t , gives

$$\frac{1}{t} \log \frac{\pi_t^F(x)}{\pi_t^F(x^*)} = \frac{1}{t} \log \frac{\pi_0^F(x)}{\pi_0^F(x^*)} + \frac{1}{t} \sum_{\tau=1}^t \log \frac{P_x(Y_\tau)}{P^*(Y_\tau)}.$$

Under the true state x^* , the signals $(Y_\tau)_{\tau \geq 1}$ are independently and identically distributed according to P^* . Moreover, because Y is finite and both P_x and P^* have full support, the random variable

$$\log \frac{P_x(Y_\tau)}{P^*(Y_\tau)}$$

takes only finitely many finite values and is therefore bounded and integrable. The strong law of large numbers consequently implies

$$\frac{1}{t} \sum_{\tau=1}^t \log \frac{P_x(Y_\tau)}{P^*(Y_\tau)} \longrightarrow \sum_{y \in Y} P^*(y) \log \frac{P_x(y)}{P^*(y)} = -K(x) \quad \mathbb{P}^*\text{-a.s.},$$

where the equality follows from the definition of K in (1). The initial log-odds term is a finite constant divided by t , so it converges to zero. Therefore,

$$\frac{1}{t} \log \frac{\pi_t^F(x)}{\pi_t^F(x^*)} \longrightarrow -K(x) < 0 \quad \mathbb{P}^*\text{-a.s.} \quad (25)$$

Because the true state is identified, $K(x) > 0$ for every $x \neq x^*$. Fix such an x . On the probability-one event on which (25) holds, there is a finite random date T_x such that, for every $t \geq T_x$,

$$\frac{1}{t} \log \frac{\pi_t^F(x)}{\pi_t^F(x^*)} \leq -\frac{K(x)}{2}.$$

Equivalently,

$$0 \leq \frac{\pi_t^F(x)}{\pi_t^F(x^*)} \leq \exp \left\{ -\frac{tK(x)}{2} \right\} \longrightarrow 0.$$

Thus $\pi_t^F(x)/\pi_t^F(x^*) \rightarrow 0$ almost surely for every $x \neq x^*$. Since X is finite,

$$\pi_t^F(x^*) = \frac{1}{1 + \sum_{x \neq x^*} \pi_t^F(x)/\pi_t^F(x^*)} \longrightarrow 1 \quad \mathbb{P}^*\text{-a.s.}$$

Therefore $\pi_t^F \rightarrow \delta_{(a^*, b^*)}$. Marginalization consists of finite sums and is continuous on $\overline{\Delta}(X)$, so $\mathbf{M}(\pi_t^F) \rightarrow (\delta_{a^*}, \delta_{b^*})$ almost surely. $\blacksquare$

Proof of Theorem 1

We first record the log-odds identities used below.

Lemma A.1. *For every $q_t = (\alpha_t, \beta_t) \in Q$, every distinct $a, a' \in A$, and every distinct $b, b' \in B$,*

$$\log \frac{\alpha_{t+1}(a)}{\alpha_{t+1}(a')} = \log \frac{\alpha_t(a)}{\alpha_t(a')} + \log \frac{\tilde{P}_A(Y_{t+1} \mid a; \beta_t)}{\tilde{P}_A(Y_{t+1} \mid a'; \beta_t)}, \quad (26)$$

$$\log \frac{\beta_{t+1}(b)}{\beta_{t+1}(b')} = \log \frac{\beta_t(b)}{\beta_t(b')} + \log \frac{\tilde{P}_B(Y_{t+1} \mid b; \alpha_t)}{\tilde{P}_B(Y_{t+1} \mid b'; \alpha_t)}. \quad (27)$$

We also use a bounded martingale law of large numbers.

Lemma A.2. *Let $(\xi_t)_{t \geq 1}$ be martingale differences with respect to $(\mathcal{F}_t)$. If $|\xi_t| \leq L$ almost surely for every t , then $t^{-1} \sum_{s=1}^t \xi_s \rightarrow 0$ almost surely.*

For part (i), suppose first that weak coordinate-wise KL minimality fails along the A -coordinate. Then there exists $a \neq \bar{a}$ such that $K(\bar{a}, \bar{b}) > K(a, \bar{b})$. For this alternative a ,

define

$$\Phi_a(\beta) := \sum_{y \in Y} P^*(y) \log \frac{\tilde{P}_A(y \mid \bar{a}; \beta)}{\tilde{P}_A(y \mid a; \beta)}. \quad (28)$$

Recall from (21) that $\tilde{P}_A(y \mid a; \beta) = \sum_{b \in B} \beta(b) P_{ab}(y)$. At the pure marginal $\delta_{\bar{b}}$, the effective likelihoods reduce to the primitive likelihoods:

$$\tilde{P}_A(y \mid \bar{a}; \delta_{\bar{b}}) = P_{\bar{a}\bar{b}}(y), \quad \tilde{P}_A(y \mid a; \delta_{\bar{b}}) = P_{a\bar{b}}(y).$$

Hence

$$\Phi_a(\delta_{\bar{b}}) = \sum_{y \in Y} P^*(y) \log \frac{P_{\bar{a}\bar{b}}(y)}{P_{a\bar{b}}(y)} = K(a, \bar{b}) - K(\bar{a}, \bar{b}) < 0. \quad (29)$$

Because all primitive likelihoods have full support, every effective likelihood is strictly positive. Since Y is finite, the drift function in (28) is continuous on the closed simplex $\overline{\Delta}(B)$. It follows from (29) that there exist a neighborhood N of $\delta_{\bar{b}}$ and a constant $\kappa > 0$ such that

$$\Phi_a(\beta) \leq -\kappa \quad \text{for every } \beta \in N.$$

Fix $q_0 \in Q$, and consider the event $E := \{q_t \rightarrow (\delta_{\bar{a}}, \delta_{\bar{b}})\}$. Write $q_t = (\alpha_t, \beta_t)$ and define the log odds of $\bar{a}$ against the particular alternative a by

$$Z_t^a := \log \frac{\alpha_t(\bar{a})}{\alpha_t(a)}.$$

All finite-date marginal beliefs have full support, so Z_t^a is finite at every t .

If E occurs, then $\beta_t \rightarrow \delta_{\bar{b}}$. Since N is a neighborhood of $\delta_{\bar{b}}$, there is a finite random date T such that $\beta_t \in N$ for every $t \geq T$. Consequently, $\Phi_a(\beta_t) \leq -\kappa$ for every $t \geq T$ on E . At the same time, convergence on E implies $\alpha_t(\bar{a}) \rightarrow 1$ and $\alpha_t(a) \rightarrow 0$. Hence

$$\frac{\alpha_t(\bar{a})}{\alpha_t(a)} \rightarrow +\infty, \quad Z_t^a \rightarrow +\infty \quad (30)$$

on E .

We now show that (30) is impossible. Applying (26) with $a = \bar{a}$ and $a' = a$ gives

$$Z_{t+1}^a - Z_t^a = \log \frac{\tilde{P}_A(Y_{t+1} \mid \bar{a}; \beta_t)}{\tilde{P}_A(Y_{t+1} \mid a; \beta_t)}.$$

Because Y_{t+1} is independently distributed according to P^* conditional on $\mathcal{F}_t$, while β_t is

$\mathcal{F}_t$ -measurable,

$$\mathbb{E}^*[Z_{t+1}^a - Z_t^a \mid \mathcal{F}_t] = \Phi_a(\beta_t).$$

Define the centered increment $\xi_{t+1}^a := Z_{t+1}^a - Z_t^a - \Phi_a(\beta_t)$ and the associated martingale

$$M_t^a := \sum_{s=0}^{t-1} \xi_{s+1}^a.$$

Indeed, ξ_{t+1}^a is $\mathcal{F}_{t+1}$ -measurable and satisfies $\mathbb{E}^*[\xi_{t+1}^a \mid \mathcal{F}_t] = 0$. Summing the identity defining ξ_{t+1}^a from $s = 0$ to $t - 1$ gives

$$Z_t^a = Z_0^a + \sum_{s=0}^{t-1} \Phi_a(\beta_s) + M_t^a. \quad (31)$$

Since A , B , and Y are finite and every primitive likelihood has full support, there exists

$$\underline{p} := \min_{(a', b') \in X} \min_{y \in Y} P_{a'b'}(y) > 0.$$

Also,

$$\underline{p} \leq \tilde{P}_A(y \mid a'; \beta) \leq 1$$

for every $a' \in A$, $y \in Y$, and $\beta \in \overline{\Delta}(B)$. Therefore,

$$\left| \log \frac{\tilde{P}_A(y \mid \bar{a}; \beta)}{\tilde{P}_A(y \mid a; \beta)} \right| \leq -\log \underline{p}.$$

The same bound applies to its expectation $\Phi_a(\beta)$. Thus the centered increments ξ_{t+1}^a are uniformly bounded. Define the probability-one martingale-LLN event

$$\Omega_0 := \left\{ \frac{M_t^a}{t} \rightarrow 0 \right\}.$$

Lemma A.2 gives $\mathbb{P}^*(\Omega_0) = 1$. We conduct the remainder of the contradiction on $E \cap \Omega_0$.

On $E \cap \Omega_0$, let T be the finite random date defined above. For every $t > T$,

$$\sum_{s=0}^{t-1} \Phi_a(\beta_s) = \sum_{s=0}^{T-1} \Phi_a(\beta_s) + \sum_{s=T}^{t-1} \Phi_a(\beta_s) \leq \sum_{s=0}^{T-1} \Phi_a(\beta_s) - \kappa(t - T).$$

Dividing by t ,

$$\frac{1}{t} \sum_{s=0}^{t-1} \Phi_a(\beta_s) \leq \frac{1}{t} \sum_{s=0}^{T-1} \Phi_a(\beta_s) - \kappa \frac{t-T}{t}.$$

The first term on the right converges to zero because it is a finite path-dependent constant divided by t , while $(t-T)/t \rightarrow 1$. Hence

$$\limsup_{t \rightarrow \infty} \frac{1}{t} \sum_{s=0}^{t-1} \Phi_a(\beta_s) \leq -\kappa \quad \text{on } E \cap \Omega_0.$$

Dividing the decomposition in (31) by t gives

$$\frac{Z_t^a}{t} = \frac{Z_0^a}{t} + \frac{1}{t} \sum_{s=0}^{t-1} \Phi_a(\beta_s) + \frac{M_t^a}{t}.$$

On $E \cap \Omega_0$, the first and third terms converge to zero. Therefore,

$$\limsup_{t \rightarrow \infty} \frac{Z_t^a}{t} \leq -\kappa,$$

which implies $Z_t^a \rightarrow -\infty$. This contradicts (30), which holds on E . It follows that $\mathbb{P}^*(E \cap \Omega_0) = 0$. Since $\mathbb{P}^*(\Omega_0) = 1$, we have $\mathbb{P}^*(E) = 0$. Thus the weak coordinate-wise KL inequality cannot fail along the A -coordinate if convergence to $(\delta_{\bar{a}}, \delta_{\bar{b}})$ has positive probability. The argument for the B -coordinate is symmetric.

For part (ii), we use the following stronger uniform result.

Proposition A.1. *Let $\mathcal{K} \subset \mathcal{P}$ be compact and fix $\bar{x} = (\bar{a}, \bar{b})$. Suppose*

$$\kappa_0 := \inf_{P \in \mathcal{K}} \min \left\{ \min_{a \neq \bar{a}} [K^P(a, \bar{b}) - K^P(\bar{a}, \bar{b})], \min_{b \neq \bar{b}} [K^P(\bar{a}, b) - K^P(\bar{a}, \bar{b})] \right\} > 0.$$

Then there exist $C > 0$ and $\gamma > 0$ such that, with $d := |A| + |B| - 2$, for every $P \in \mathcal{K}$, every $r > 0$, and every $q_0 \in \mathcal{G}_{C+r}(\bar{x})$,

$$\mathbb{P}_{q_0}^P (q_t \in \mathcal{G}_C(\bar{x}) \text{ for every } t \geq 0 \text{ and } q_t \rightarrow (\delta_{\bar{a}}, \delta_{\bar{b}})) \geq 1 - de^{-\gamma r}.$$

Apply Proposition A.1 to the singleton $\mathcal{K} = \{P\}$. Strict coordinate-wise KL minimality gives its positive-gap condition. Fix $\varepsilon \in (0, 1)$, choose $r > 0$ such that $de^{-\gamma r} \leq \varepsilon$, and choose

$\eta_\varepsilon \in (0, 1)$ sufficiently close to one that

$$\log \frac{\eta_\varepsilon}{1 - \eta_\varepsilon} \geq C + r.$$

If $\alpha_0(\bar{a}) \geq \eta_\varepsilon$, then $\alpha_0(a) \leq 1 - \eta_\varepsilon$ for every $a \neq \bar{a}$, so

$$\log \frac{\alpha_0(\bar{a})}{\alpha_0(a)} \geq \log \frac{\eta_\varepsilon}{1 - \eta_\varepsilon} \geq C + r.$$

The same argument applies to β_0 . By the definition of the reinforcing region in (22), $q_0 \in \mathcal{G}_{C+r}(\bar{x})$. Proposition A.1 therefore gives

$$\mathbb{P}^*(q_t \rightarrow (\delta_{\bar{a}}, \delta_{\bar{b}})) \geq 1 - de^{-\gamma r} \geq 1 - \varepsilon.$$

This proves part (ii). ■

Proof of Corollary 1

Suppose first that (i) holds. Then $\bar{x}$ is a strict coordinate-wise KL minimum, so Theorem 1(ii) directly gives (ii).

Suppose next that (ii) holds. Set $\varepsilon = 1/2$, and let $\eta_{1/2} \in (0, 1)$ be the corresponding threshold. Define

$$\alpha_0(\bar{a}) := \frac{1 + \eta_{1/2}}{2}, \quad \alpha_0(a) := \frac{1 - \alpha_0(\bar{a})}{|A| - 1} \quad \text{for every } a \neq \bar{a},$$

and, analogously,

$$\beta_0(\bar{b}) := \frac{1 + \eta_{1/2}}{2}, \quad \beta_0(b) := \frac{1 - \beta_0(\bar{b})}{|B| - 1} \quad \text{for every } b \neq \bar{b}.$$

Because $\eta_{1/2} < 1$, every coordinate of α_0 and β_0 is strictly positive. Hence $q_0 = (\alpha_0, \beta_0) \in Q$. Moreover,

$$\alpha_0(\bar{a}) > \eta_{1/2}, \quad \beta_0(\bar{b}) > \eta_{1/2}.$$

Applying (ii) to this initial record gives

$$\mathbb{P}^*(q_t \rightarrow (\delta_{\bar{a}}, \delta_{\bar{b}})) \geq \frac{1}{2} > 0.$$

Thus (iii) holds.

Finally, suppose that (iii) holds. Then there exists $q_0 \in Q$ such that

$$\mathbb{P}^*(q_t \rightarrow (\delta_{\bar{a}}, \delta_{\bar{b}})) > 0.$$

Theorem 1(i) therefore implies that $\bar{x}$ is a weak coordinate-wise KL minimum. By the assumed absence of coordinate-wise KL ties, none of the inequalities defining weak coordinate-wise KL minimality can hold with equality. They must therefore all be strict, so $\bar{x}$ is a strict coordinate-wise KL minimum. Hence (i) holds. $\blacksquare$

Proof of Theorem 2

For $y \in \{0, 1\}$ and an integer $n \geq 0$, let y^n denote n consecutive occurrences of y , with y^0 the empty word. Recall that $x^* = (L, s)$, $\bar{x} = (H, g)$, and

$$P_{Hg}^0(1) = \frac{3}{5}, \quad P_{Hs}^0(1) = \frac{1}{10}, \quad P_{Lg}^0(1) = \frac{9}{10}, \quad P_{Ls}^0(1) = \frac{1}{2}.$$

For $\zeta > 0$, recall from (13) that

$$Q_\zeta := \left\{ (\alpha, \beta) \in Q : \left| \log \frac{\alpha(H)}{\alpha(L)} \right| \leq \zeta, \left| \log \frac{\beta(g)}{\beta(s)} \right| \leq \zeta \right\}.$$

We use the following two auxiliary results.

Proposition A.2. *There is an open neighborhood $\mathcal{O}_0$ of P^0 , with compact closure contained in $\mathcal{P}$, such that the following holds for every $\varepsilon \in (0, 1)$ and $C > 0$.⁹*

- (i) *There are an integer $n_{\bar{x}}$ and $\rho_{\bar{x}} > 0$ such that, for every $P \in \mathcal{O}_0$ and $q = (\alpha, \beta) \in Q$ with $\alpha(H)\beta(g) \geq \varepsilon$,*

$$U_{1^{n_{\bar{x}}}0^{\lfloor n_{\bar{x}}/3 \rfloor}}^P(q) \in \mathcal{G}_C(\bar{x}), \quad \mathbb{P}^P((Y_1, \dots, Y_{n_{\bar{x}} + \lfloor n_{\bar{x}}/3 \rfloor}) = 1^{n_{\bar{x}}}0^{\lfloor n_{\bar{x}}/3 \rfloor}) \geq \rho_{\bar{x}}.$$

- (ii) *There are an integer n_{x^*} and $\rho_{x^*} > 0$ such that, for every $P \in \mathcal{O}_0$ and $q = (\alpha, \beta) \in Q$ with $\alpha(L)\beta(s) \geq \varepsilon$,*

$$U_{0^{n_{x^*}}1^{\lfloor 2n_{x^*}/5 \rfloor}}^P(q) \in \mathcal{G}_C(x^*), \quad \mathbb{P}^P((Y_1, \dots, Y_{n_{x^*} + \lfloor 2n_{x^*}/5 \rfloor}) = 0^{n_{x^*}}1^{\lfloor 2n_{x^*}/5 \rfloor}) \geq \rho_{x^*}.$$

The integers and probability bounds do not depend on P or the initial record.

⁹For a real number z , $\lfloor z \rfloor$ is the greatest integer not exceeding z , and $\lceil z \rceil$ is the least integer not smaller than z .

Theorem A.1. Fix $P \in \mathcal{P}$, let $A = \{a_0, a_1\}$ and $B = \{b_0, b_1\}$, and define $x^0 := (a_0, b_0)$ and $x^1 := (a_1, b_1)$. If both diagonal states are strict coordinate-wise KL minima and are finitely accessible away from zero product mass, then, for every $q_0 \in Q$,

$$\mathbb{P}_{q_0}^P(q_t \rightarrow (\delta_{a_0}, \delta_{b_0}) \text{ or } q_t \rightarrow (\delta_{a_1}, \delta_{b_1})) = 1,$$

and both limiting events have strictly positive probability.

Step 1 (A common neighborhood and uniform gaps). Let $\mathcal{O}_0$ be the neighborhood supplied by Proposition A.2. At P^0 , the true state (L, s) is identified and the false state (H, g) is a strict coordinate-wise KL minimum. Since each $K^P(a, b)$ is continuous in P , we may choose a single open neighborhood $\mathcal{O}$ of P^0 such that $\overline{\mathcal{O}} \subset \mathcal{O}_0$, $\overline{\mathcal{O}}$ is compact, and

$$\kappa_{\text{id}} := \min_{P \in \overline{\mathcal{O}}} \min\{K^P(H, g), K^P(H, s), K^P(L, g)\} > 0, \quad (32)$$

$$\kappa_{\bar{x}} := \min_{P \in \overline{\mathcal{O}}} \min\{K^P(L, g) - K^P(H, g), K^P(H, s) - K^P(H, g)\} > 0. \quad (33)$$

Since $K^P(L, s) = 0$ for every P , (32) identifies (L, s) throughout $\overline{\mathcal{O}}$, and hence makes it a strict coordinate-wise KL minimum. Equation (33) makes (H, g) a strict coordinate-wise KL minimum throughout $\overline{\mathcal{O}}$.

Step 2 (Full Bayesian consistency). For part (i), fix $P \in \mathcal{O}$ and $q_0 \in Q$. The initial joint belief $\pi_0^F = R(q_0)$ has full support, and (L, s) is identified under P by (32). Proposition 2 therefore gives

$$\pi_t^F \rightarrow \delta_{(L, s)} \quad \mathbb{P}^P\text{-a.s.}$$

Step 3 (Exhaustive diagonal selection). For part (ii), fix $P \in \mathcal{O}$ and $q_0 \in Q$. As established above, both (L, s) and (H, g) are strict coordinate-wise KL minima. Proposition A.2 implies that both are finitely accessible away from zero product mass. Hence Theorem A.1, applied with $a_0 = L$, $a_1 = H$, $b_0 = s$, and $b_1 = g$, yields

$$\mathbb{P}^P(q_t \rightarrow (\delta_L, \delta_s) \text{ or } q_t \rightarrow (\delta_H, \delta_g)) = 1.$$

The same theorem gives

$$\mathbb{P}^P(q_t \rightarrow (\delta_L, \delta_s)) > 0, \quad \mathbb{P}^P(q_t \rightarrow (\delta_H, \delta_g)) > 0.$$

Step 4 (A uniform false-convergence lower bound). It remains to prove the uniform lower bound on false convergence. By (33), Proposition A.1, applied to the compact set

$\overline{\mathcal{O}}$ and $\bar{x} = (H, g)$, gives constants $C_{\bar{x}}, \gamma_{\bar{x}} > 0$ such that, for every $P \in \overline{\mathcal{O}}$, $r > 0$, and $q \in \mathcal{G}_{C_{\bar{x}}+r}(\bar{x})$,

$$\mathbb{P}_q^P(q_t \rightarrow (\delta_H, \delta_g)) \geq 1 - 2e^{-\gamma_{\bar{x}}r}. \quad (34)$$

Fix $\zeta > 0$. If $q_0 = (\alpha, \beta) \in Q_\zeta$, the bounded log odds imply

$$\alpha(H), \alpha(L), \beta(g), \beta(s) \geq \frac{1}{1 + e^\zeta}.$$

Recalling the product mass in (23), we obtain

$$m_{\bar{x}}(q_0) = \alpha(H)\beta(g) \geq \varepsilon_\zeta := \frac{1}{(1 + e^\zeta)^2}.$$

Choose $r_\zeta > 0$ such that $1 - 2e^{-\gamma_{\bar{x}}r_\zeta} \geq 1/2$. By (34),

$$q \in \mathcal{G}_{C_{\bar{x}}+r_\zeta}(\bar{x}) \implies \mathbb{P}_q^P(q_t \rightarrow (\delta_H, \delta_g)) \geq \frac{1}{2}$$

uniformly over $P \in \overline{\mathcal{O}}$.

Apply Proposition A.2(i) with $\varepsilon = \varepsilon_\zeta$ and $C = C_{\bar{x}} + r_\zeta$. There are an integer n_ζ and $\rho_\zeta > 0$, independent of $P \in \mathcal{O}$ and $q_0 \in Q_\zeta$, such that, for

$$w_\zeta := 1^{n_\zeta} 0^{\lfloor n_\zeta/3 \rfloor}, \quad T_\zeta := n_\zeta + \lfloor n_\zeta/3 \rfloor,$$

we have

$$U_{w_\zeta}^P(q_0) \in \mathcal{G}_{C_{\bar{x}}+r_\zeta}(\bar{x}), \quad \mathbb{P}^P((Y_1, \dots, Y_{T_\zeta}) = w_\zeta) \geq \rho_\zeta.$$

Let

$$A_\zeta := \{(Y_1, \dots, Y_{T_\zeta}) = w_\zeta\}, \quad E_{\bar{x}} := \{q_t \rightarrow (\delta_H, \delta_g)\}.$$

On A_ζ , we have $q_{T_\zeta} \in \mathcal{G}_{C_{\bar{x}}+r_\zeta}(\bar{x})$. Since the recursive process is Markov and future signals are independent of $\mathcal{F}_{T_\zeta}$,

$$\mathbb{P}^P(E_{\bar{x}} \mid \mathcal{F}_{T_\zeta}) \geq \frac{1}{2} \quad \text{on } A_\zeta.$$

Because $A_\zeta \in \mathcal{F}_{T_\zeta}$,

$$\mathbb{P}^P(E_{\bar{x}}) \geq \mathbb{E}^P[\mathbf{1}_{A_\zeta} \mathbb{P}^P(E_{\bar{x}} \mid \mathcal{F}_{T_\zeta})] \geq \frac{1}{2} \mathbb{P}^P(A_\zeta) \geq \frac{\rho_\zeta}{2}.$$

Thus, setting $\eta_\zeta := \rho_\zeta/2 > 0$,

$$\inf_{P \in \mathcal{O}} \inf_{q_0 \in Q_\zeta} \mathbb{P}^P(q_t \rightarrow (\delta_H, \delta_g)) \geq \eta_\zeta.$$

This completes the proof. ■

Proof of Theorem 3

We prove the implications

$$(i) \implies (ii) \implies (iii) \implies (i).$$

(i) **implies** (ii). Suppose the likelihood system has no coordinate interaction. Fix $q = (\alpha, \beta) \in Q$ and $y \in Y$. By (15), there exist strictly positive functions f_A^y and f_B^y such that

$$P_{ab}(y) = f_A^y(a) f_B^y(b)$$

for every $a \in A$ and $b \in B$. Since $R(q)(a, b) = \alpha(a)\beta(b)$, Bayes' rule gives

$$\mathbb{B}_y(R(q))(a, b) = \frac{\alpha(a)\beta(b)f_A^y(a)f_B^y(b)}{\sum_{a' \in A} \sum_{b' \in B} \alpha(a')\beta(b')f_A^y(a')f_B^y(b')}.$$

The denominator separates across the two coordinates:

$$\sum_{a' \in A} \sum_{b' \in B} \alpha(a')\beta(b')f_A^y(a')f_B^y(b') = \left(\sum_{a' \in A} \alpha(a')f_A^y(a') \right) \left(\sum_{b' \in B} \beta(b')f_B^y(b') \right).$$

Therefore,

$$\mathbb{B}_y(R(q))(a, b) = \frac{\alpha(a)f_A^y(a)}{\sum_{a' \in A} \alpha(a')f_A^y(a')} \frac{\beta(b)f_B^y(b)}{\sum_{b' \in B} \beta(b')f_B^y(b')}.$$

Define

$$\alpha_y(a) := \frac{\alpha(a)f_A^y(a)}{\sum_{a' \in A} \alpha(a')f_A^y(a')}, \quad \beta_y(b) := \frac{\beta(b)f_B^y(b)}{\sum_{b' \in B} \beta(b')f_B^y(b')}.$$

Then $\mathbb{B}_y(R(q))(a, b) = \alpha_y(a)\beta_y(b)$, so the posterior is a product belief. This proves (i) $\implies$ (ii).

(ii) **implies** (iii). Suppose every Bayesian update from a product belief remains a product belief. Fix a finite signal history $(y_1, \dots, y_T)$. We prove by induction that

$$R(q_t) = \pi_t^F \quad \text{for every } t = 0, \dots, T.$$

The equality holds at $t = 0$ by construction. Suppose it holds at some date $t < T$, so that

$\pi_t^F = R(q_t)$. After observing y_{t+1} , the full Bayesian learner forms

$$\pi_{t+1}^F = \mathbf{B}_{y_{t+1}}(\pi_t^F) = \mathbf{B}_{y_{t+1}}(R(q_t)).$$

By statement (ii), this posterior is a product belief. Any product belief equals the product reconstruction of its own marginals, so, using (2),

$$\mathbf{B}_{y_{t+1}}(R(q_t)) = R(\mathbf{M}(\mathbf{B}_{y_{t+1}}(R(q_t)))) .$$

By the recursive dynamics in (7), $q_{t+1} = \mathbf{M}(\mathbf{B}_{y_{t+1}}(R(q_t)))$. Combining the preceding equalities gives $\pi_{t+1}^F = R(q_{t+1})$. The induction step is complete. Hence $R(q_t) = \pi_t^F$ at every date along every finite signal history. This proves (ii) $\Rightarrow$ (iii).

(iii) **implies** (i). Suppose statement (iii) holds. Fix an arbitrary signal $y \in Y$. We show that the likelihoods associated with y are multiplicatively separable across the two coordinates.¹⁰

Let $q^u = (\alpha^u, \beta^u) \in Q$ be the uniform marginal record,

$$\alpha^u(a) := \frac{1}{|A|}, \quad \beta^u(b) := \frac{1}{|B|}.$$

Its product reconstruction is the uniform joint belief,

$$R(q^u)(a, b) = \frac{1}{|A||B|}.$$

Initialize the full Bayesian learner at $\pi_0^F = R(q^u)$ and consider the one-signal history consisting only of y . Bayes' rule gives

$$\pi_1^F(a, b) = \frac{P_{ab}(y)}{\sum_{a' \in A} \sum_{b' \in B} P_{a'b'}(y)}.$$

By statement (iii), $\pi_1^F = R(q_1)$, so π_1^F is a product belief. Writing $q_1 = (\alpha_1, \beta_1)$, we have

$$\pi_1^F(a, b) = \alpha_1(a)\beta_1(b) \quad \text{for every } a \in A, b \in B.$$

¹⁰Let L_y denote the positive matrix with entries $L_y(a, b) := P_{ab}(y)$. The no-coordinate-interaction condition is equivalent to L_y having rank one.

Consequently, for every $a, a' \in A$ and $b, b' \in B$,

$$\pi_1^F(a, b)\pi_1^F(a', b') = \pi_1^F(a, b')\pi_1^F(a', b). \quad (35)$$

Indeed, after substituting $\pi_1^F(a, b) = \alpha_1(a)\beta_1(b)$, both sides of (35) equal

$$\alpha_1(a)\alpha_1(a')\beta_1(b)\beta_1(b').$$

To translate (35) into a restriction on the likelihoods, define the normalizing constant

$$D_y := \sum_{\tilde{a} \in A} \sum_{\tilde{b} \in B} P_{\tilde{a}\tilde{b}}(y).$$

Because every likelihood is strictly positive, $D_y > 0$. Under the uniform product prior,

$$\pi_1^F(a, b) = \frac{P_{ab}(y)}{D_y}.$$

Substituting this expression into (35), the common positive normalizing factor D_y^{-2} cancels from both sides, yielding

$$P_{ab}(y)P_{a'b'}(y) = P_{ab'}(y)P_{a'b}(y) \quad \text{for every } a, a' \in A, b, b' \in B. \quad (36)$$

Hence every 2×2 minor of the likelihood matrix associated with y vanishes.

It remains to show that (36) implies the factorization in (15). Fix arbitrary reference coordinates $a_0 \in A$ and $b_0 \in B$, and define

$$f_A^y(a) := P_{ab_0}(y), \quad f_B^y(b) := \frac{P_{a_0b}(y)}{P_{a_0b_0}(y)}.$$

Full support implies $P_{a_0b_0}(y) > 0$, so both functions are well defined and strictly positive. Applying (36) with $a' = a_0$ and $b' = b_0$ gives

$$P_{ab}(y)P_{a_0b_0}(y) = P_{ab_0}(y)P_{a_0b}(y).$$

Dividing by $P_{a_0b_0}(y)$,

$$P_{ab}(y) = P_{ab_0}(y) \frac{P_{a_0b}(y)}{P_{a_0b_0}(y)} = f_A^y(a)f_B^y(b).$$

Since $y \in Y$ was arbitrary, this factorization holds for every signal. Hence the likelihood system has no coordinate interaction. This proves (iii) $\Rightarrow$ (i), completing the equivalence.

Finally, we establish the additive KL decomposition stated after the theorem. Under the no-interaction condition, write

$$P_{ab}(y) = f_A^y(a)f_B^y(b),$$

and define

$$\begin{aligned} c &:= \sum_{y \in Y} P^*(y) \log P^*(y), \\ \kappa_A(a) &:= - \sum_{y \in Y} P^*(y) \log f_A^y(a), \\ \kappa_B(b) &:= - \sum_{y \in Y} P^*(y) \log f_B^y(b). \end{aligned}$$

Then

$$\begin{aligned} K(a, b) &= \sum_{y \in Y} P^*(y) \log \frac{P^*(y)}{P_{ab}(y)} \\ &= \sum_{y \in Y} P^*(y) \log P^*(y) - \sum_{y \in Y} P^*(y) \log f_A^y(a) - \sum_{y \in Y} P^*(y) \log f_B^y(b) \\ &= c + \kappa_A(a) + \kappa_B(b). \end{aligned}$$

This establishes (16).

Suppose $(\bar{a}, \bar{b})$ is a weak coordinate-wise KL minimum. For every $a \neq \bar{a}$, $K(\bar{a}, \bar{b}) \leq K(a, \bar{b})$, and (16) therefore gives $\kappa_A(\bar{a}) \leq \kappa_A(a)$. Hence $\bar{a}$ minimizes κ_A over A . Similarly, $K(\bar{a}, \bar{b}) \leq K(\bar{a}, b)$ for every $b \neq \bar{b}$ implies that $\bar{b}$ minimizes κ_B over B . Consequently, for every $(a, b) \in X$,

$$K(\bar{a}, \bar{b}) = c + \kappa_A(\bar{a}) + \kappa_B(\bar{b}) \leq c + \kappa_A(a) + \kappa_B(b) = K(a, b).$$

Thus every weak coordinate-wise KL minimum is a global KL minimum. Under identification, x^* is the unique global KL minimum, so it is also the unique weak coordinate-wise KL minimum. ■

Proof of Corollary 2

Let $p_A := \pi_A(a_1)$, $p_B := \pi_B(b_1)$, and $z := \pi(a_1, b_1)$. The remaining cell probabilities are $p_A - z$, $p_B - z$, and $1 - p_A - p_B + z$,¹¹ so full support is equivalent to

$$\max\{0, p_A + p_B - 1\} < z < \min\{p_A, p_B\}. \quad (37)$$

¹¹For example, $p_A = \pi(a_1, b_0) + \pi(a_1, b_1)$, so $\pi(a_1, b_0) = p_A - z$.

For fixed marginals, the interaction coordinate in (17) can be written as

$$\ell(\pi) = f_{p_A, p_B}(z) := \log \frac{z(1 - p_A - p_B + z)}{(p_A - z)(p_B - z)}.$$

On the feasible interval in (37),

$$f'_{p_A, p_B}(z) = \frac{1}{z} + \frac{1}{1 - p_A - p_B + z} + \frac{1}{p_A - z} + \frac{1}{p_B - z} > 0.$$

Moreover, $f_{p_A, p_B}(z) \rightarrow -\infty$ at the lower endpoint and $f_{p_A, p_B}(z) \rightarrow +\infty$ at the upper endpoint of (37). Hence, for fixed marginals, each interaction value determines a unique z . The marginals then determine the other three cell probabilities. Thus $(\mathbf{M}(\pi), \ell(\pi))$ uniquely determines π .

Let

$$D_y(\pi) := \sum_{a \in A} \sum_{b \in B} \pi(a, b) P_{ab}(y).$$

Substituting the Bayesian update in (3) into the posterior cross-ratio gives

$$\begin{aligned} \exp(\ell(\mathbf{B}_y(\pi))) &= \frac{\pi(a_1, b_1) P_{a_1 b_1}(y) \pi(a_0, b_0) P_{a_0 b_0}(y)}{\pi(a_1, b_0) P_{a_1 b_0}(y) \pi(a_0, b_1) P_{a_0 b_1}(y)} \\ &= \frac{\pi(a_1, b_1) \pi(a_0, b_0)}{\pi(a_1, b_0) \pi(a_0, b_1)} \frac{P_{a_1 b_1}(y) P_{a_0 b_0}(y)}{P_{a_1 b_0}(y) P_{a_0 b_1}(y)} \\ &= \exp(\ell(\pi) + \lambda_y), \end{aligned}$$

where the common factor $D_y(\pi)^{-2}$ cancels and λ_y is defined in (18). Taking logarithms gives

$$\ell(\mathbf{B}_y(\pi)) = \ell(\pi) + \lambda_y.$$

Moreover, $R(\mathbf{M}(\pi))$ is a product belief by (2), so its cross-ratio is one and $\ell(R(\mathbf{M}(\pi))) = 0$. Together, these are the identities in (19).

If $\pi_0^F = R(q_0)$, then $\ell(\pi_0^F) = 0$. Repeated application of (19) therefore gives

$$\ell(\pi_t^F) = \sum_{\tau=1}^t \lambda_{Y_\tau},$$

which is (20). The uniqueness argument above shows that $(\mathbf{M}(\pi_t^F), \ell(\pi_t^F))$ determines π_t^F at every date. More explicitly, a learner who reconstructs the unique joint belief with the stored marginals and interaction, updates it by Bayes' rule, and then stores the new marginals together with $\ell_{t+1} = \ell_t + \lambda_{Y_{t+1}}$ reproduces the full Bayesian posterior by induction.

Finally, $\lambda_y = 0$ for every y if and only if every binary likelihood matrix has cross-ratio one, or equivalently satisfies (36). Such a positive binary matrix has rank one. Theorem 3 then gives universal sufficiency of the unaugmented marginal record, and the converse follows from the same theorem. ■

A.2 Auxiliary Results and Proofs

Proof of Lemma A.1

Starting from $R(q_t)$, the updated A -marginal is

$$\alpha_{t+1}(a) = \frac{\alpha_t(a) \sum_{b \in B} \beta_t(b) P_{ab}(Y_{t+1})}{\sum_{\tilde{a} \in A} \alpha_t(\tilde{a}) \sum_{b \in B} \beta_t(b) P_{\tilde{a}b}(Y_{t+1})}.$$

Taking the ratio of this expression for a and a' , cancelling the common denominator, using the effective likelihoods in (21), and taking logarithms gives (26). The same calculation after summing the joint posterior over A gives (27). ■

Proof of Lemma A.2

Let $M_t := \sum_{s=1}^t \xi_s$. Azuma–Hoeffding’s inequality gives,¹² for every $\varepsilon > 0$,

$$\mathbb{P}(|M_t| > \varepsilon t) \leq 2 \exp\left(-\frac{\varepsilon^2 t}{2L^2}\right).$$

The right-hand side is summable in t , so the first Borel–Cantelli lemma implies that $|M_t| > \varepsilon t$ occurs only finitely often almost surely.¹³ Applying this conclusion to $\varepsilon = 1/n$, $n \geq 1$, gives $M_t/t \rightarrow 0$ almost surely. ■

Lemma A.3. *Let $\mathcal{K} \subset \mathcal{P}$ be compact.*

- (i) *For every $y \in Y$, $(P, q) \mapsto U_y^P(q)$ is continuous on $\mathcal{K} \times \overline{Q}$.*
- (ii) *Every pure record is fixed:*

$$U_y^P(\delta_a, \delta_b) = (\delta_a, \delta_b)$$

for every $P \in \mathcal{K}$, $y \in Y$, and $(a, b) \in X$.

¹²Azuma–Hoeffding’s inequality states that if (M_t) is a martingale and $|M_s - M_{s-1}| \leq c_s$ almost surely, then

$$\mathbb{P}(|M_t - M_0| \geq a) \leq 2 \exp\left\{-\frac{a^2}{2 \sum_{s=1}^t c_s^2}\right\}.$$

Here $M_0 = 0$, $c_s = L$, and $a = \varepsilon t$.

¹³The first Borel–Cantelli lemma states that if $\sum_t \mathbb{P}(A_t) < \infty$, then A_t occurs infinitely often with probability zero.

(iii) For every $y \in Y$, every distinct $a, a' \in A$, and every distinct $b, b' \in B$, the maps

$$(P, \beta) \mapsto \log \frac{\tilde{P}_A^P(y | a; \beta)}{\tilde{P}_A^P(y | a'; \beta)}, \quad (P, \alpha) \mapsto \log \frac{\tilde{P}_B^P(y | b; \alpha)}{\tilde{P}_B^P(y | b'; \alpha)}$$

are jointly continuous and bounded in absolute value by a common finite constant on the corresponding compact domains.¹⁴

(iv) For a fixed candidate $\bar{x} = (\bar{a}, \bar{b})$, define

$$\Phi_a^P(\beta) := \sum_{y \in Y} P_{a^*b^*}(y) \log \frac{\tilde{P}_A^P(y | \bar{a}; \beta)}{\tilde{P}_A^P(y | a; \beta)}, \quad a \neq \bar{a}, \quad (38)$$

$$\Psi_b^P(\alpha) := \sum_{y \in Y} P_{a^*b^*}(y) \log \frac{\tilde{P}_B^P(y | \bar{b}; \alpha)}{\tilde{P}_B^P(y | b; \alpha)}, \quad b \neq \bar{b}. \quad (39)$$

The functions in (38) and (39) are jointly continuous in (P, β) and (P, α) , respectively.

Proof of Lemma A.3

Compact containment in the full-support region gives

$$\underline{p}_{\mathcal{K}} := \min_{P \in \mathcal{K}} \min_{(a,b) \in X} \min_{y \in Y} P_{ab}(y) > 0.$$

Thus every effective likelihood in (21) lies in $[\underline{p}_{\mathcal{K}}, 1]$. The denominator in every Bayesian update is also at least $\underline{p}_{\mathcal{K}}$, including on $\overline{\mathcal{Q}}$. The update maps are therefore ratios of continuous functions with uniformly positive denominators, proving (i).

If $q = (\delta_a, \delta_b)$, then $R(q) = \delta_{(a,b)}$, and Bayesian updating preserves this degenerate prior. Taking marginals proves (ii). The effective log-likelihood ratios are continuous and bounded in absolute value by $-\log \underline{p}_{\mathcal{K}}$, proving (iii). Finally, each drift in (38) and (39) is a finite sum of products of continuous functions, proving (iv). $\blacksquare$

Lemma A.4. Let $Z_t^1, \dots, Z_t^d$ be real-valued processes adapted to $(\mathcal{F}_t)$,¹⁵ with increments $\xi_{t+1}^j := Z_{t+1}^j - Z_t^j$. Fix $C \in \mathbb{R}$, and define

$$\tau := \inf \left\{ t \geq 0 : \min_{1 \leq j \leq d} Z_t^j < C \right\}.$$

Suppose there are constants $L < \infty$ and $\kappa > 0$ such that, for every j , $|\xi_{t+1}^j| \leq L$ almost

¹⁴A function $f(x, z)$ is jointly continuous if $x_n \rightarrow x$ and $z_n \rightarrow z$ together imply $f(x_n, z_n) \rightarrow f(x, z)$.

¹⁵A stochastic process $(Z_t)_{t \geq 0}$ is adapted to a filtration $(\mathcal{F}_t)_{t \geq 0}$ if Z_t is $\mathcal{F}_t$ -measurable for every t . Thus, at each date t , the value Z_t is determined by the information contained in $\mathcal{F}_t$.

surely and

$$\mathbb{E}[\xi_{t+1}^j \mid \mathcal{F}_t] \geq \kappa \quad \text{on } \{t < \tau\}.$$

Then there exists $\gamma > 0$, depending only on L and κ , such that, for every $r > 0$,

$$\min_{1 \leq j \leq d} Z_0^j \geq C + r \implies \mathbb{P}(\tau < \infty) \leq de^{-\gamma r}.$$

On $\{\tau = \infty\}$,

$$\liminf_{t \rightarrow \infty} \frac{Z_t^j}{t} \geq \kappa \quad \text{for every } j.$$

Proof of Lemma A.4

Because each Z_t^j is adapted, τ is a stopping time with respect to $(\mathcal{F}_t)$. Choose $\theta > 0$ sufficiently small that

$$\frac{\theta L^2 e^{\theta L}}{2} \leq \kappa.$$

For every $|\xi| \leq L$, Taylor's formula gives

$$e^{-\theta \xi} \leq 1 - \theta \xi + \frac{\theta^2 L^2 e^{\theta L}}{2}.$$

Hence, on $\{t < \tau\}$,

$$\begin{aligned} \mathbb{E}[e^{-\theta \xi_{t+1}^j} \mid \mathcal{F}_t] &\leq 1 - \theta \mathbb{E}[\xi_{t+1}^j \mid \mathcal{F}_t] + \frac{\theta^2 L^2 e^{\theta L}}{2} \\ &\leq 1 - \theta \kappa + \frac{\theta^2 L^2 e^{\theta L}}{2} \leq 1. \end{aligned}$$

Thus

$$\mathbb{E}[e^{-\theta \xi_{t+1}^j} \mid \mathcal{F}_t] \leq 1 \quad \text{on } \{t < \tau\}. \quad (40)$$

For each j , define

$$S_t^j := \exp \left\{ -\theta \left(Z_{\min\{t, \tau\}}^j - Z_0^j \right) \right\}.$$

The process is nonnegative and adapted because $Z_{\min\{t, \tau\}}^j$ is $\mathcal{F}_t$ -measurable. Moreover, the bounded-increment assumption gives

$$|Z_{\min\{t, \tau\}}^j - Z_0^j| \leq tL,$$

so $0 \leq S_t^j \leq e^{\theta t L} < \infty$, and $S_0^j = 1$.

We next verify the supermartingale property. On $\{t \geq \tau\}$, the process has already been

stopped and $S_{t+1}^j = S_t^j$. On $\{t < \tau\}$, integer-valuedness of τ implies $\tau \geq t + 1$, so

$$S_{t+1}^j = S_t^j e^{-\theta \xi_{t+1}^j}.$$

Since $\{t < \tau\}$ and $\{t \geq \tau\}$ are $\mathcal{F}_t$ -measurable, (40) gives

$$\begin{aligned} \mathbb{E}[S_{t+1}^j \mid \mathcal{F}_t] &= \mathbf{1}_{\{t \geq \tau\}} S_t^j + \mathbf{1}_{\{t < \tau\}} S_t^j \mathbb{E}[e^{-\theta \xi_{t+1}^j} \mid \mathcal{F}_t] \\ &\leq \mathbf{1}_{\{t \geq \tau\}} S_t^j + \mathbf{1}_{\{t < \tau\}} S_t^j = S_t^j. \end{aligned}$$

Hence $(S_t^j)_{t \geq 0}$ is a nonnegative supermartingale with $S_0^j = 1$.

Suppose now that $\min_j Z_0^j \geq C + r$. Fix j . On $\{\tau < \infty, Z_\tau^j < C\}$,

$$Z_\tau^j - Z_0^j < C - (C + r) = -r,$$

so $S_\tau^j > e^{\theta r}$. Therefore,

$$\{\tau < \infty, Z_\tau^j < C\} \subseteq \left\{ \sup_{t \geq 0} S_t^j \geq e^{\theta r} \right\}.$$

Ville's inequality yields¹⁶

$$\mathbb{P}(\tau < \infty, Z_\tau^j < C) \leq e^{-\theta r}.$$

If $\tau < \infty$, then at least one of the d coordinates satisfies $Z_\tau^j < C$. Hence, by the union bound,

$$\mathbb{P}(\tau < \infty) \leq d e^{-\theta r}. \quad (41)$$

Set $\gamma := \theta$.

It remains to establish the growth conclusion on $\{\tau = \infty\}$. Telescoping the increments and adding and subtracting their conditional expectations gives

$$Z_t^j = Z_0^j + \sum_{s=0}^{t-1} \mathbb{E}[\xi_{s+1}^j \mid \mathcal{F}_s] + M_t^j, \quad M_t^j := \sum_{s=0}^{t-1} (\xi_{s+1}^j - \mathbb{E}[\xi_{s+1}^j \mid \mathcal{F}_s]). \quad (42)$$

¹⁶Ville's inequality states that if $(S_t)_{t \geq 0}$ is a nonnegative supermartingale, then, for every $a > 0$,

$$\mathbb{P}\left(\sup_{t \geq 0} S_t \geq a\right) \leq \frac{\mathbb{E}[S_0]}{a}.$$

Here $S_0^j = 1$ and $a = e^{\theta r}$.

The process M_t^j is a martingale. Moreover,

$$|\mathbb{E}[\xi_{t+1}^j \mid \mathcal{F}_t]| \leq \mathbb{E}[|\xi_{t+1}^j| \mid \mathcal{F}_t] \leq L,$$

so $|M_{t+1}^j - M_t^j| \leq 2L$. Lemma A.2 therefore gives $M_t^j/t \rightarrow 0$ almost surely.

On $\{\tau = \infty\}$, we have $s < \tau$ for every finite s , and hence $\mathbb{E}[\xi_{s+1}^j \mid \mathcal{F}_s] \geq \kappa$. Dividing (42) by t therefore gives

$$\frac{Z_t^j}{t} \geq \frac{Z_0^j}{t} + \kappa + \frac{M_t^j}{t}.$$

Since $Z_0^j/t \rightarrow 0$ and $M_t^j/t \rightarrow 0$,

$$\liminf_{t \rightarrow \infty} \frac{Z_t^j}{t} \geq \kappa \quad \text{almost surely on } \{\tau = \infty\}.$$

Because $\kappa > 0$, this also implies $Z_t^j \rightarrow +\infty$ on $\{\tau = \infty\}$. ■

Proof of Proposition A.1

For $a \neq \bar{a}$ and $b \neq \bar{b}$, define

$$Z_t^a := \log \frac{\alpha_t(\bar{a})}{\alpha_t(a)}, \quad W_t^b := \log \frac{\beta_t(\bar{b})}{\beta_t(b)}.$$

Recall the drift functions Φ_a^P and Ψ_b^P from (38) and (39). At the pure anchors, the effective likelihoods reduce to the corresponding primitive likelihoods, so

$$\Phi_a^P(\delta_{\bar{b}}) = K^P(a, \bar{b}) - K^P(\bar{a}, \bar{b}), \quad \Psi_b^P(\delta_{\bar{a}}) = K^P(\bar{a}, b) - K^P(\bar{a}, \bar{b}). \quad (43)$$

By the assumption of Proposition A.1, every quantity in (43) is at least $\kappa_0 > 0$, uniformly over $P \in \mathcal{K}$ and all relevant coordinate deviations. Joint continuity from Lemma A.3, compactness of $\mathcal{K}$, and finiteness of A and B therefore give neighborhoods N_A of $\delta_{\bar{a}}$ and N_B of $\delta_{\bar{b}}$ such that

$$\Phi_a^P(\beta) \geq \frac{\kappa_0}{2} \quad \text{for } P \in \mathcal{K}, \beta \in N_B, a \neq \bar{a}, \quad \Psi_b^P(\alpha) \geq \frac{\kappa_0}{2} \quad \text{for } P \in \mathcal{K}, \alpha \in N_A, b \neq \bar{b}. \quad (44)$$

We next choose the reinforcing region so that (44) applies throughout it. If

$$\min_{a \neq \bar{a}} \log \frac{\alpha(\bar{a})}{\alpha(a)} \geq C,$$

then $\alpha(a) \leq e^{-C}\alpha(\bar{a})$ for every $a \neq \bar{a}$. Summing these inequalities gives

$$1 = \alpha(\bar{a}) + \sum_{a \neq \bar{a}} \alpha(a) \leq \alpha(\bar{a}) [1 + (|A| - 1)e^{-C}],$$

and therefore

$$\alpha(\bar{a}) \geq \frac{1}{1 + (|A| - 1)e^{-C}}. \quad (45)$$

The lower bound in (45) converges to one as $C \rightarrow \infty$. Hence there exists $C_A < \infty$ such that the preceding log-odds bound with $C = C_A$ implies $\alpha \in N_A$. The same argument gives $C_B < \infty$ such that

$$\min_{b \neq \bar{b}} \log \frac{\beta(\bar{b})}{\beta(b)} \geq C_B$$

implies $\beta \in N_B$. Choose $C \geq \max\{C_A, C_B\}$. By (22),

$$\mathcal{G}_C(\bar{x}) \subseteq N_A \times N_B.$$

Define the first exit time

$$\tau := \inf\{t \geq 0 : q_t \notin \mathcal{G}_C(\bar{x})\} = \inf\left\{t \geq 0 : \min\left\{\min_{a \neq \bar{a}} Z_t^a, \min_{b \neq \bar{b}} W_t^b\right\} < C\right\}.$$

On $\{t < \tau\}$, we have $\alpha_t \in N_A$ and $\beta_t \in N_B$. The log-odds identities (26) and (27) give

$$\mathbb{E}^P[Z_{t+1}^a - Z_t^a \mid \mathcal{F}_t] = \Phi_a^P(\beta_t), \quad \mathbb{E}^P[W_{t+1}^b - W_t^b \mid \mathcal{F}_t] = \Psi_b^P(\alpha_t).$$

Hence, by (44), every one of the $d := |A| + |B| - 2$ log-odds processes has conditional drift at least $\kappa_0/2$ before τ .

Lemma A.3(iii) also gives a constant $L < \infty$, independent of $P \in \mathcal{K}$, such that

$$|Z_{t+1}^a - Z_t^a| \leq L \quad \text{for every } a \neq \bar{a}, \quad |W_{t+1}^b - W_t^b| \leq L \quad \text{for every } b \neq \bar{b}.$$

Thus all d log-odds processes satisfy the hypotheses of Lemma A.4 with common drift lower bound $\kappa_0/2$, common increment bound L , and boundary level C .

Fix $r > 0$ and $q_0 \in \mathcal{G}_{C+r}(\bar{x})$. By (22),

$$Z_0^a \geq C + r \quad \text{for every } a \neq \bar{a}, \quad W_0^b \geq C + r \quad \text{for every } b \neq \bar{b}.$$

Lemma A.4 therefore gives a constant $\gamma > 0$, depending only on $\kappa_0/2$ and L , such that

$$\mathbb{P}_{q_0}^P(\tau < \infty) \leq de^{-\gamma r}. \quad (46)$$

Because κ_0 and L are uniform over $P \in \mathcal{K}$, the same γ works for every $P \in \mathcal{K}$.

On $\{\tau = \infty\}$, Lemma A.4 gives

$$\liminf_{t \rightarrow \infty} \frac{Z_t^a}{t} \geq \frac{\kappa_0}{2} \quad \text{for every } a \neq \bar{a}, \quad \liminf_{t \rightarrow \infty} \frac{W_t^b}{t} \geq \frac{\kappa_0}{2} \quad \text{for every } b \neq \bar{b}.$$

Hence $Z_t^a \rightarrow +\infty$ and $W_t^b \rightarrow +\infty$ for every alternative coordinate. Consequently,

$$\frac{\alpha_t(a)}{\alpha_t(\bar{a})} = e^{-Z_t^a} \rightarrow 0, \quad \frac{\beta_t(b)}{\beta_t(\bar{b})} = e^{-W_t^b} \rightarrow 0.$$

Since the marginals sum to one,

$$\alpha_t(\bar{a}) = \frac{1}{1 + \sum_{a \neq \bar{a}} e^{-Z_t^a}} \rightarrow 1, \quad \beta_t(\bar{b}) = \frac{1}{1 + \sum_{b \neq \bar{b}} e^{-W_t^b}} \rightarrow 1.$$

Because A and B are finite, this implies $\alpha_t \rightarrow \delta_{\bar{a}}$, $\beta_t \rightarrow \delta_{\bar{b}}$, and therefore

$$q_t \rightarrow (\delta_{\bar{a}}, \delta_{\bar{b}}) \quad \text{on } \{\tau = \infty\}.$$

Finally, $\{\tau = \infty\}$ is exactly the event that $q_t \in \mathcal{G}_C(\bar{x})$ for every $t \geq 0$, and convergence to $(\delta_{\bar{a}}, \delta_{\bar{b}})$ holds on this event. Therefore, by (46),

$$\mathbb{P}_{q_0}^P(q_t \in \mathcal{G}_C(\bar{x}) \text{ for every } t \geq 0 \text{ and } q_t \rightarrow (\delta_{\bar{a}}, \delta_{\bar{b}})) \geq 1 - de^{-\gamma r},$$

uniformly over $P \in \mathcal{K}$. ■

Proof of Proposition A.2

For a binary record $q = (\alpha, \beta)$, write

$$u := \log \frac{\alpha(H)}{\alpha(L)}, \quad v := \log \frac{\beta(g)}{\beta(s)}, \quad \sigma(z) := \frac{e^z}{1 + e^z}. \quad (47)$$

Since $\alpha(H) = \sigma(u)$ and $\beta(g) = \sigma(v)$, the log-odds identities (26) and (27) give

$$u_{t+1} = u_t + g_u^P(Y_{t+1}, v_t), \quad v_{t+1} = v_t + g_v^P(Y_{t+1}, u_t), \quad (48)$$

where

$$\begin{aligned} g_u^P(y, v) &:= \log \frac{\sigma(v)P_{Hg}(y) + (1 - \sigma(v))P_{Hs}(y)}{\sigma(v)P_{Lg}(y) + (1 - \sigma(v))P_{Ls}(y)}, \\ g_v^P(y, u) &:= \log \frac{\sigma(u)P_{Hg}(y) + (1 - \sigma(u))P_{Lg}(y)}{\sigma(u)P_{Hs}(y) + (1 - \sigma(u))P_{Ls}(y)}. \end{aligned} \quad (49)$$

Step 1 (Compactifying the log-odds increments). To make uniformity over the unbounded log-odds coordinates explicit, define, for $r, s \in [0, 1]$,

$$\widehat{g}_u^P(y, r) := \log \frac{rP_{Hg}(y) + (1 - r)P_{Hs}(y)}{rP_{Lg}(y) + (1 - r)P_{Ls}(y)},$$

and

$$\widehat{g}_v^P(y, s) := \log \frac{sP_{Hg}(y) + (1 - s)P_{Lg}(y)}{sP_{Hs}(y) + (1 - s)P_{Ls}(y)}.$$

Then

$$g_u^P(y, v) = \widehat{g}_u^P(y, \sigma(v)), \quad g_v^P(y, u) = \widehat{g}_v^P(y, \sigma(u)).$$

Full support makes every denominator strictly positive. Hence these compactified increment functions are jointly continuous in the likelihood system and the mixing weight on $\mathcal{P} \times [0, 1]$. The unbounded domains for u and v therefore create no uniformity problem: they enter only through weights in the compact interval $[0, 1]$.

Step 2 (Uniform steering bounds for both diagonal states). We first construct a finite signal word that steers the record toward $\bar{x} = (H, g)$. At the baseline P^0 , put $s := \sigma(u) \in [0, 1]$. Direct substitution into (49) gives

$$e^{g_v^{P^0}(1, u)} = \frac{9 - 3s}{5 - 4s}, \quad e^{g_v^{P^0}(0, u)} = \frac{1 + 3s}{5 + 4s}. \quad (50)$$

Both ratios in (50) are increasing in s , and hence

$$\inf_u g_v^{P^0}(1, u) = \log \frac{9}{5}, \quad \inf_u g_v^{P^0}(0, u) = -\log 5.$$

Moreover, as $v \rightarrow +\infty$, $\sigma(v) \rightarrow 1$, so

$$\lim_{v \rightarrow +\infty} g_u^{P^0}(1, v) = -\log \frac{3}{2}, \quad \lim_{v \rightarrow +\infty} g_u^{P^0}(0, v) = \log 4.$$

The strict inequalities

$$\log \frac{9}{5} > \frac{\log 5}{3}, \quad \frac{\log 4}{3} > \log \frac{3}{2}$$

allow us to choose positive constants a_g, e_g, b_H, c_H such that

$$a_g < \log \frac{9}{5}, \quad e_g > \log 5, \quad b_H > \log \frac{3}{2}, \quad c_H < \log 4, \quad a_g > \frac{e_g}{3}, \quad \frac{c_H}{3} > b_H.$$

The first two baseline bounds have strict slack relative to a_g and $-e_g$. Joint continuity of $\widehat{g}_v^P$ on the compact weight interval $[0, 1]$ therefore gives an open neighborhood $\mathcal{N}_g$ of P^0 , with compact closure contained in $\mathcal{P}$, such that

$$\widehat{g}_v^P(1, s) \geq a_g, \quad \widehat{g}_v^P(0, s) \geq -e_g$$

for every $P \in \mathcal{N}_g$ and every $s \in [0, 1]$. Moreover,

$$\widehat{g}_u^{P^0}(1, 1) = -\log \frac{3}{2} > -b_H, \quad \widehat{g}_u^{P^0}(0, 1) = \log 4 > c_H.$$

By joint continuity at mixing weight one, after shrinking $\mathcal{N}_g$ if necessary, there exists $r_g \in (0, 1)$ such that

$$\widehat{g}_u^P(1, r) \geq -b_H, \quad \widehat{g}_u^P(0, r) \geq c_H$$

for every $P \in \mathcal{N}_g$ and every $r \in [r_g, 1]$. Set

$$V_g := \log \frac{r_g}{1 - r_g}.$$

Because $\sigma(v) \geq r_g$ whenever $v \geq V_g$, we obtain, uniformly over $P \in \mathcal{N}_g$,

$$\begin{aligned} g_v^P(1, u) &\geq a_g, & g_v^P(0, u) &\geq -e_g & \text{for every } u, \\ g_u^P(1, v) &\geq -b_H, & g_u^P(0, v) &\geq c_H & \text{whenever } v \geq V_g. \end{aligned} \tag{51}$$

We also need analogous bounds for steering toward $x^* = (L, s)$. Define

$$\bar{u} := -u = \log \frac{\alpha(L)}{\alpha(H)}, \quad \bar{v} := -v = \log \frac{\beta(s)}{\beta(g)}.$$

By (48),

$$\bar{u}_{t+1} - \bar{u}_t = -g_u^P(Y_{t+1}, -\bar{v}_t), \quad \bar{v}_{t+1} - \bar{v}_t = -g_v^P(Y_{t+1}, -\bar{u}_t).$$

At P^0 ,

$$\inf_{\bar{u}} \{-g_v^{P^0}(0, -\bar{u})\} = \log \frac{9}{4}, \quad \inf_{\bar{u}} \{-g_v^{P^0}(1, -\bar{u})\} = -\log 6,$$

and

$$\lim_{\bar{v} \rightarrow +\infty} \{-g_u^{P^0}(0, -\bar{v})\} = -\log \frac{9}{5}, \quad \lim_{\bar{v} \rightarrow +\infty} \{-g_u^{P^0}(1, -\bar{v})\} = \log 5.$$

Since

$$\log \frac{9}{4} > \frac{2 \log 6}{5}, \quad \frac{2 \log 5}{5} > \log \frac{9}{5},$$

we can choose positive constants a_s, e_s, b_L, c_L such that

$$a_s < \log \frac{9}{4}, \quad e_s > \log 6, \quad b_L > \log \frac{9}{5}, \quad c_L < \log 5, \quad a_s > \frac{2e_s}{5}, \quad \frac{2c_L}{5} > b_L.$$

Write $\bar{s} := \sigma(\bar{u})$ and $\bar{r} := \sigma(\bar{v})$. Because $\sigma(-\bar{u}) = 1 - \bar{s}$ and $\sigma(-\bar{v}) = 1 - \bar{r}$, the quantities $-g_v^P(y, -\bar{u})$ and $-g_u^P(y, -\bar{v})$ are jointly continuous functions of $(P, \bar{s})$ and $(P, \bar{r})$, respectively, on compact weight intervals. The baseline inequalities have strict slack, so the same compactness argument gives an open neighborhood $\mathcal{N}_s$ of P^0 , with compact closure contained in $\mathcal{P}$, and a weight $r_s \in (0, 1)$ such that the first pair of bounds below holds for every $\bar{s} \in [0, 1]$, while the second pair holds whenever $\bar{r} \in [r_s, 1]$. Set

$$V_s := \log \frac{r_s}{1 - r_s}.$$

Then, uniformly over $P \in \mathcal{N}_s$,

$$\begin{aligned} -g_v^P(0, -\bar{u}) &\geq a_s, & -g_v^P(1, -\bar{u}) &\geq -e_s \quad \text{for every } \bar{u}, \\ -g_u^P(0, -\bar{v}) &\geq -b_L, & -g_u^P(1, -\bar{v}) &\geq c_L \quad \text{whenever } \bar{v} \geq V_s. \end{aligned} \tag{52}$$

Choose an open neighborhood $\mathcal{O}_0$ of P^0 such that

$$\overline{\mathcal{O}_0} \subset \mathcal{N}_g \cap \mathcal{N}_s.$$

The closure $\overline{\mathcal{O}_0}$ is compact and contained in $\mathcal{P}$, and both (51) and (52) hold throughout $\overline{\mathcal{O}_0}$. Lemma A.3 then gives a finite constant L_{acc} bounding the absolute value of every one-period change in $u, v, \bar{u}, \bar{v}$, uniformly over $P \in \overline{\mathcal{O}_0}$ and all records.

Step 3 (Steering toward the false diagonal). Fix $\varepsilon \in (0, 1)$ and $C > 0$. We first prove part (i). Suppose $q_0 = (\alpha_0, \beta_0)$ satisfies $\alpha_0(H)\beta_0(g) \geq \varepsilon$. Since $\alpha_0(H), \beta_0(g) \leq 1$, this implies

$$\alpha_0(H) \geq \varepsilon, \quad \beta_0(g) \geq \varepsilon.$$

Since also $\alpha_0(L), \beta_0(s) \leq 1$, the log-odds coordinates in (47) satisfy

$$u_0 \geq \log \varepsilon, \quad v_0 \geq \log \varepsilon.$$

Consider the first block of n consecutive signals equal to 1. By (51),

$$v_{t+1} - v_t = g_v^P(1, u_t) \geq a_g$$

at every date in this block. Telescoping gives

$$v_t = v_0 + \sum_{s=0}^{t-1} g_v^P(1, u_s) \geq v_0 + a_g t \geq \log \varepsilon + a_g t, \quad 0 \leq t \leq n.$$

Let

$$N_g(\varepsilon) := \max \left\{ 0, \left\lceil \frac{V_g - \log \varepsilon}{a_g} \right\rceil \right\}, \quad D_g(\varepsilon) := L_{\text{acc}} N_g(\varepsilon) - \log \varepsilon.$$

Then $v_t \geq V_g$ for every $N_g(\varepsilon) \leq t \leq n$ during the first block. Before reaching this region, the uniform increment bound implies that u can decrease by at most L_{acc} per period. Afterward, (51) implies that each remaining signal 1 changes u by at least $-b_H$. Hence, for every $n \geq N_g(\varepsilon)$,

$$u_n \geq u_0 - L_{\text{acc}} N_g(\varepsilon) - b_H(n - N_g(\varepsilon)).$$

Using $u_0 \geq \log \varepsilon$ and discarding the nonnegative term $b_H N_g(\varepsilon)$, we obtain

$$u_n \geq -b_H n - D_g(\varepsilon).$$

Now append $m_n := \lfloor n/3 \rfloor$ zeros. By (51), each zero lowers v by at most e_g . Thus, for every $0 \leq k \leq m_n$,

$$v_{n+k} \geq \log \varepsilon + a_g n - e_g k \geq \log \varepsilon + \left(a_g - \frac{e_g}{3} \right) n.$$

Since $a_g - e_g/3 > 0$,

$$\log \varepsilon + \left(a_g - \frac{e_g}{3} \right) n \rightarrow +\infty \quad \text{as } n \rightarrow \infty.$$

Hence, for all sufficiently large n , we have $v_{n+k} \geq V_g$ for every $0 \leq k \leq m_n$. Equation (51) then implies that each zero in the second block increases u by at least c_H . Since $m_n = \lfloor n/3 \rfloor \geq n/3 - 1$,

$$u_{n+m_n} \geq u_n + c_H m_n \geq \left(\frac{c_H}{3} - b_H \right) n - D_g(\varepsilon) - c_H, \quad v_{n+m_n} \geq \log \varepsilon + \left(a_g - \frac{e_g}{3} \right) n.$$

Because $c_H/3 - b_H > 0$ and $a_g - e_g/3 > 0$, both lower bounds tend to $+\infty$. We may therefore choose a finite integer $n_{\bar{x}}$, depending only on ε and C , such that both final log odds are at least C , uniformly over $P \in \mathcal{O}_0$ and all initial records satisfying $\alpha(H)\beta(g) \geq \varepsilon$. In the binary case, the definition in (22) reduces to

$$q \in \mathcal{G}_C(H, g) \iff u \geq C \text{ and } v \geq C.$$

It follows that

$$U_{1_{n_{\bar{x}}0}^{[n_{\bar{x}}/3]}}^P(q_0) \in \mathcal{G}_C(\bar{x}).$$

Step 4 (Steering toward the true diagonal). We now prove part (ii) using $(\bar{u}, \bar{v})$ and the word $0^{n_1}1^{k_n}$, where $k_n := \lfloor 2n/5 \rfloor$. Suppose $\alpha_0(L)\beta_0(s) \geq \varepsilon$. As above, this implies

$$\bar{u}_0 \geq \log \varepsilon, \quad \bar{v}_0 \geq \log \varepsilon.$$

Consider first a block of n consecutive zeros. By (52),

$$\bar{v}_{t+1} - \bar{v}_t = -g_v^P(0, -\bar{u}_t) \geq a_s$$

at every date in this block. Hence

$$\bar{v}_t = \bar{v}_0 + \sum_{j=0}^{t-1} [-g_v^P(0, -\bar{u}_j)] \geq \bar{v}_0 + a_s t \geq \log \varepsilon + a_s t, \quad 0 \leq t \leq n.$$

Define

$$N_s(\varepsilon) := \max \left\{ 0, \left\lceil \frac{V_s - \log \varepsilon}{a_s} \right\rceil \right\}, \quad D_s(\varepsilon) := L_{\text{acc}} N_s(\varepsilon) - \log \varepsilon.$$

Then $\bar{v}_t \geq V_s$ for every $N_s(\varepsilon) \leq t \leq n$ during the first block. Before that date, the uniform increment bound allows $\bar{u}$ to decrease by at most L_{acc} per period. Thereafter, each remaining zero changes $\bar{u}$ by at least $-b_L$. Thus, for every $n \geq N_s(\varepsilon)$,

$$\bar{u}_n \geq \bar{u}_0 - L_{\text{acc}} N_s(\varepsilon) - b_L (n - N_s(\varepsilon)).$$

Using $\bar{u}_0 \geq \log \varepsilon$ and discarding the nonnegative term $b_L N_s(\varepsilon)$, we obtain

$$\bar{u}_n \geq -b_L n - D_s(\varepsilon).$$

Append $k_n = \lfloor 2n/5 \rfloor$ consecutive ones. By (52), each one lowers $\bar{v}$ by at most e_s . Therefore,

for every $0 \leq k \leq k_n$,

$$\bar{v}_{n+k} \geq \log \varepsilon + a_s n - e_s k \geq \log \varepsilon + \left(a_s - \frac{2e_s}{5}\right) n.$$

Since $a_s - 2e_s/5 > 0$, this lower bound tends to $+\infty$. For all sufficiently large n , it is therefore at least V_s for every $0 \leq k \leq k_n$. Equation (52) then implies that each one in the second block increases $\bar{u}$ by at least c_L . Since $k_n \geq 2n/5 - 1$, we obtain

$$\bar{u}_{n+k_n} \geq \bar{u}_n + c_L k_n \geq \left(\frac{2c_L}{5} - b_L\right) n - D_s(\varepsilon) - c_L,$$

and

$$\bar{v}_{n+k_n} \geq \log \varepsilon + \left(a_s - \frac{2e_s}{5}\right) n.$$

Both lower bounds diverge to $+\infty$ because $a_s > 2e_s/5$ and $2c_L/5 > b_L$. We may therefore choose a finite integer n_{x^*} , depending only on ε and C , such that both final log odds are at least C , uniformly over $P \in \mathcal{O}_0$ and all initial records satisfying $\alpha(L)\beta(s) \geq \varepsilon$. In the binary case,

$$q \in \mathcal{G}_C(L, s) \iff \bar{u} \geq C \text{ and } \bar{v} \geq C.$$

Consequently,

$$U_{0^{n_{x^*}+1}^{2n_{x^*}/5}}^P(q_0) \in \mathcal{G}_C(x^*).$$

Step 5 (Uniform probabilities of the steering words). It remains to bound the probabilities of the two steering words. Since $\overline{\mathcal{O}_0}$ is compact and contained in the full-support likelihood space,

$$\underline{p} := \min_{P \in \overline{\mathcal{O}_0}} \min\{P_{Ls}(0), P_{Ls}(1)\} > 0.$$

Under the true state (L, s) , the signals are independently and identically distributed according to P_{Ls} . Thus every binary word $(y_1, \dots, y_T)$ has probability

$$\mathbb{P}^P(Y_1 = y_1, \dots, Y_T = y_T) = \prod_{t=1}^T P_{Ls}(y_t) \geq \underline{p}^T$$

uniformly over $P \in \mathcal{O}_0$. Therefore we may take

$$\rho_{\bar{x}} := \underline{p}^{n_{\bar{x}} + \lceil n_{\bar{x}}/3 \rceil}, \quad \rho_{x^*} := \underline{p}^{n_{x^*} + \lceil 2n_{x^*}/5 \rceil}.$$

The integers $n_{\bar{x}}$ and n_{x^*} , and hence the probability bounds $\rho_{\bar{x}}$ and ρ_{x^*} , depend only on ε , C , and the uniform constants fixed by $\mathcal{O}_0$, not on P or the initial record. This completes the proof. $\blacksquare$

Proposition A.3. *Fix $P \in \mathcal{P}$, and let $x = (a, b)$ be a strict coordinate-wise KL minimum that is finitely accessible away from zero product mass. Define*

$$E_x := \{q_t \rightarrow (\delta_a, \delta_b)\}, \quad h_x^P(q) := \mathbb{P}_q^P(E_x).$$

For every $q_0 \in Q$,

$$h_x^P(q_t) = \mathbb{P}_{q_0}^P(E_x \mid \mathcal{F}_t) \longrightarrow \mathbf{1}_{E_x} \quad \mathbb{P}_{q_0}^P\text{-a.s.}$$

Moreover, for every $\varepsilon \in (0, 1)$, there is $c_{x,\varepsilon}^P > 0$ such that $m_x(q) \geq \varepsilon$ implies $h_x^P(q) \geq c_{x,\varepsilon}^P$ for every $q \in Q$. Consequently, $m_x(q_t) \rightarrow 0$ almost surely on E_x^c .

Proof of Proposition A.3

The recursive process is a time-homogeneous Markov chain driven by independent signals, so

$$h_x^P(q_t) = \mathbb{E}_{q_0}^P[\mathbf{1}_{E_x} \mid \mathcal{F}_t].$$

Since $E_x \in \mathcal{F}_\infty := \sigma(\bigcup_{t \geq 0} \mathcal{F}_t)$, Levy's upward theorem gives¹⁷

$$h_x^P(q_t) \rightarrow \mathbf{1}_{E_x} \quad \mathbb{P}_{q_0}^P\text{-a.s.}$$

Apply Proposition A.1 to the singleton likelihood family $\{P\}$. The proposition gives constants $C, \gamma > 0$ such that, with $d := |A| + |B| - 2$, every record $q' \in \mathcal{G}_{C+r}(x)$ satisfies

$$\mathbb{P}_{q'}^P(q_t \rightarrow (\delta_a, \delta_b)) \geq 1 - de^{-\gamma r}$$

for every $r > 0$. Choose $r > 0$ large enough that $\theta_x := 1 - de^{-\gamma r} > 0$. Thus, once the recursive record enters $\mathcal{G}_{C+r}(x)$, its probability of subsequently converging to x is at least θ_x .

Now fix $\varepsilon \in (0, 1)$. Recall from (23) that $m_x(q) = \alpha(a)\beta(b)$. Since x is finitely accessible away from zero product mass, there exists a finite signal word $w = (y_1, \dots, y_T)$ such that

$$U_w^P(q) \in \mathcal{G}_{C+r}(x)$$

for every $q \in Q$ satisfying $m_x(q) \geq \varepsilon$. Thus, whenever the initial product mass on x is at

¹⁷Levy's upward theorem states that if $(\mathcal{F}_t)$ increases to $\mathcal{F}_\infty$ and Z is integrable, then $\mathbb{E}[Z \mid \mathcal{F}_t] \rightarrow \mathbb{E}[Z \mid \mathcal{F}_\infty]$ almost surely and in L^1 .

least ε , observing w moves the recursive record into the reinforcing region from which local attraction applies.

Because the true signal distribution $P_{a^*b^*}$ has full support and the signals are independently and identically distributed, the probability of observing this particular finite word is

$$p_w^P := \mathbb{P}^P((Y_1, \dots, Y_T) = w) = \prod_{s=1}^T P_{a^*b^*}(y_s) > 0.$$

This probability depends on P and w , but not on the initial record q .

Fix any $q \in Q$ with $m_x(q) \geq \varepsilon$, and let $A_w := \{(Y_1, \dots, Y_T) = w\}$. On A_w , the record at date T is $q_T = U_w^P(q) \in \mathcal{G}_{C+r}(x)$. By the Markov property and the local-attraction bound,¹⁸

$$\mathbb{P}_q^P(E_x \mid A_w) \geq \theta_x.$$

Hence

$$h_x^P(q) = \mathbb{P}_q^P(E_x) \geq \mathbb{P}_q^P(E_x \cap A_w) = \mathbb{P}_q^P(A_w) \mathbb{P}_q^P(E_x \mid A_w) \geq p_w^P \theta_x.$$

Defining $c_{x,\varepsilon}^P := p_w^P \theta_x > 0$, we obtain $h_x^P(q) \geq c_{x,\varepsilon}^P$ for every $q \in Q$ satisfying $m_x(q) \geq \varepsilon$.

On E_x^c , $h_x^P(q_t) \rightarrow 0$. Suppose, toward a contradiction, that $m_x(q_t) \not\rightarrow 0$. Then there exists an integer $n \geq 1$ such that

$$m_x(q_t) \geq \frac{1}{n}$$

along infinitely many dates. At every such date,

$$h_x^P(q_t) \geq c_{x,1/n}^P > 0,$$

contradicting $h_x^P(q_t) \rightarrow 0$. Hence $m_x(q_t) \rightarrow 0$ on E_x^c . ■

For a nonempty set S in the finite-dimensional record space, write $\text{dist}(q, S) := \inf_{z \in S} \|q - z\|$, where $\|\cdot\|$ is any norm.

Lemma A.5. *Fix $P \in \mathcal{P}$. Let $A = \{a_0, a_1\}$, $B = \{b_0, b_1\}$, and define*

$$\mathcal{V}_{\text{off}} := \{(\delta_{a_0}, \delta_{b_1}), (\delta_{a_1}, \delta_{b_0})\}.$$

¹⁸The Markov property means that, conditional on the current record q_t , the future evolution of the recursive process does not depend on the earlier history. Formally, for every measurable set A of future paths,

$$\mathbb{P}_q^P((q_{t+s})_{s \geq 0} \in A \mid \mathcal{F}_t) = \mathbb{P}_{q_t}^P((q_s)_{s \geq 0} \in A) \quad \text{a.s.}$$

Here this follows because $q_{t+1} = U_{Y_{t+1}}^P(q_t)$ and, conditional on the true state, future signals are independent of $\mathcal{F}_t$.

Let $(y_t)_{t \geq 1}$ be any sequence in Y , and let $(q_t)_{t \geq 0} \subseteq \overline{Q}$ satisfy

$$q_{t+1} = U_{y_{t+1}}^P(q_t) \quad \text{for every } t \geq 0.$$

If $\text{dist}(q_t, \mathcal{V}_{\text{off}}) \rightarrow 0$, then q_t converges to one element of $\mathcal{V}_{\text{off}}$.

Proof of Lemma A.5

Write

$$z^0 := (\delta_{a_0}, \delta_{b_1}), \quad z^1 := (\delta_{a_1}, \delta_{b_0}).$$

These are the two off-diagonal vertices of $\overline{Q}$: each component of each record is a degenerate marginal distribution. They are distinct because $a_0 \neq a_1$ and $b_0 \neq b_1$. Hence $z^0 - z^1 \neq 0$, and the positive-definiteness of a norm gives $D := \|z^0 - z^1\| > 0$.

By Lemma A.3, applied to the singleton likelihood family $\{P\}$, each update map U_y^P is continuous on $\overline{Q}$ and fixes every pure record. In particular,

$$U_y^P(z^0) = z^0, \quad U_y^P(z^1) = z^1$$

for every $y \in Y$.

Choose $\eta \in (0, D/3)$. By continuity of U_y^P at z^i , for every $i \in \{0, 1\}$ and $y \in Y$, there exists $\delta_{i,y} > 0$ such that

$$\|q - z^i\| < \delta_{i,y} \implies \|U_y^P(q) - z^i\| < \eta.$$

Since Y is finite, choose

$$\delta \in \left(0, \frac{D}{3}\right)$$

smaller than every $\delta_{i,y}$. Then

$$\|q - z^i\| < \delta \implies \|U_y^P(q) - z^i\| < \eta \tag{53}$$

for every $i \in \{0, 1\}$ and $y \in Y$.

Now suppose $\text{dist}(q_t, \mathcal{V}_{\text{off}}) \rightarrow 0$. There exists a finite date T such that, for every $t \geq T$, $\text{dist}(q_t, \mathcal{V}_{\text{off}}) < \delta$. Because $2\delta < D$, the δ -neighborhoods of z^0 and z^1 are disjoint. Thus, at every $t \geq T$, the record q_t lies within distance δ of a unique vertex.

Suppose q_t lies within distance δ of z^i . By (53),

$$\|q_{t+1} - z^i\| = \|U_{y_{t+1}}^P(q_t) - z^i\| < \eta.$$

Since $t+1 \geq T$, the record q_{t+1} lies within distance δ of one of the two vertices. It cannot lie within distance δ of the other vertex z^{1-i} , because the triangle inequality would then give

$$D = \|z^0 - z^1\| \leq \|z^i - q_{t+1}\| + \|q_{t+1} - z^{1-i}\| < \eta + \delta < \frac{2D}{3},$$

a contradiction. Therefore q_{t+1} lies within distance δ of the same vertex z^i .

It follows inductively that, after date T , the nearby vertex cannot switch. Thus there exists $i \in \{0, 1\}$ such that q_t lies in the δ -neighborhood of z^i for every $t \geq T$. For these dates, the other vertex is not the nearest element of $\mathcal{V}_{\text{off}}$, so $\|q_t - z^i\| = \text{dist}(q_t, \mathcal{V}_{\text{off}}) \rightarrow 0$. Hence $q_t \rightarrow z^i$. ■

Proof of Theorem A.1

Step 1 (Positive probability of both diagonal limits). For $i \in \{0, 1\}$, let $E_i := \{q_t \rightarrow (\delta_{a_i}, \delta_{b_i})\}$. Since $q_0 \in Q$, both diagonal product masses are strictly positive. For each i , choose $\varepsilon_i \in (0, m_{x^i}(q_0)]$. Proposition A.3 then gives

$$\mathbb{P}_{q_0}^P(E_i) = h_{x^i}^P(q_0) \geq c_{x^i, \varepsilon_i}^P > 0.$$

Thus both diagonal limiting events have strictly positive probability.

Step 2 (Consequences of avoiding both diagonal limits). It remains to prove that one of them occurs almost surely. On $E_0^c \cap E_1^c$, Proposition A.3 gives

$$m_{x^0}(q_t) \rightarrow 0, \quad m_{x^1}(q_t) \rightarrow 0.$$

Write

$$p_t := \alpha_t(a_0), \quad r_t := \beta_t(b_0).$$

Recalling the product mass in (23), the preceding limits become

$$p_t r_t \rightarrow 0, \quad (1 - p_t)(1 - r_t) \rightarrow 0. \tag{54}$$

Every subsequential limit $(p, r) \in [0, 1]^2$ therefore satisfies

$$pr = 0, \quad (1 - p)(1 - r) = 0.$$

The only solutions are

$$(p, r) = (1, 0) \quad \text{and} \quad (p, r) = (0, 1),$$

which correspond to the two off-diagonal pure records in $\mathcal{V}_{\text{off}}$.

We claim that $\text{dist}(q_t, \mathcal{V}_{\text{off}}) \rightarrow 0$. Otherwise, there would exist $\varepsilon > 0$ and a subsequence (q_{t_k}) satisfying

$$\text{dist}(q_{t_k}, \mathcal{V}_{\text{off}}) \geq \varepsilon.$$

Compactness of $\overline{\mathcal{Q}}$ would give a further convergent subsequence. By (54), its limit would have to belong to $\mathcal{V}_{\text{off}}$, contradicting the distance bound. Because every realized record path satisfies $q_{t+1} = U_{Y_{t+1}}^P(q_t)$, Lemma A.5 implies that, on $E_0^c \cap E_1^c$, the recursive record converges to one of the two off-diagonal pure records.

Step 3 (Excluding the off-diagonal limits). Neither off-diagonal state is a weak coordinate-wise KL minimum. At (a_0, b_1) , changing only the second coordinate gives $x^0 = (a_0, b_0)$, and strict coordinate-wise minimality of x^0 implies

$$K^P(a_0, b_0) < K^P(a_0, b_1).$$

Similarly, $K^P(a_1, b_1) < K^P(a_1, b_0)$. Theorem 1(i) therefore assigns probability zero to convergence to either off-diagonal pure record. Hence $\mathbb{P}_{q_0}^P(E_0^c \cap E_1^c) = 0$, and therefore $\mathbb{P}_{q_0}^P(E_0 \cup E_1) = 1$. ■

B A Sufficient Condition for Point Convergence

This appendix supplements the main paper. Use its notation for the finite sets A , B , and Y , the marginal record $q_t = (\alpha_t, \beta_t)$, the closed record space $\overline{\mathcal{Q}}$, the recursive update maps U_y , the true signal law P^* , and the natural filtration $\mathcal{F}_t = \sigma(Y_1, \dots, Y_t)$. The local selection theorem in the main paper establishes convergence on a lock-in event but does not imply point convergence from every initial record. The result below gives a sufficient condition.

For $q = (\alpha, \beta) \in \overline{\mathcal{Q}}$, write $U_y(q) = (U_y^A(q), U_y^B(q))$ and define

$$\mu_a(q) := \sum_{y \in Y} P^*(y) U_y^A(q)(a) - \alpha(a), \quad \nu_b(q) := \sum_{y \in Y} P^*(y) U_y^B(q)(b) - \beta(b). \quad (55)$$

Proposition 3 (One-sided drift convergence). *Fix reference values $a_0 \in A$ and $b_0 \in B$. Suppose that, for every $a \neq a_0$, there is $\sigma_a \in \{-1, 1\}$, and, for every $b \neq b_0$, there is*

$\tau_b \in \{-1, 1\}$, such that

$$\sigma_a \mu_a(q) \geq 0, \quad \tau_b \nu_b(q) \geq 0 \quad \text{for every } q \in \overline{Q}.$$

Then, for every $q_0 \in Q$, there is a random $q_\infty \in \overline{Q}$ such that $q_t \rightarrow q_\infty$ $\mathbb{P}^*$ -almost surely.

Proof of Proposition 3

Fix $a \neq a_0$. Since q_t is $\mathcal{F}_t$ -measurable and Y_{t+1} is independently distributed according to P^* , the definition in (55) gives

$$\mathbb{E}^*[\sigma_a \alpha_{t+1}(a) \mid \mathcal{F}_t] - \sigma_a \alpha_t(a) = \sigma_a \mu_a(q_t) \geq 0.$$

Thus $(\sigma_a \alpha_t(a))_{t \geq 0}$ is a bounded submartingale and converges almost surely. Hence every $\alpha_t(a)$, $a \neq a_0$, converges, and

$$\alpha_t(a_0) = 1 - \sum_{a \neq a_0} \alpha_t(a)$$

converges as well.

The same argument applies to every $\beta_t(b)$, using $\tau_b \nu_b(q) \geq 0$. Because A and B are finite, all coordinates converge on a common probability-one event. Their limits are nonnegative and sum to one in each marginal, so they define a random $q_\infty \in \overline{Q}$. Therefore $q_t \rightarrow q_\infty$ almost surely. ■

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